

Domain-adaptive digital twins for multi-task vibration-based damage assessment of shield tunnels

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ABSTRACT

Ensuring the safety of in-service shield tunnels is vital, yet defects like voids and hidden concrete spalling often evade visual inspection. While vibration-based monitoring offers a non-intrusive alternative, its deployment is hindered by scarce damage data and high computational demands. This paper proposed a domain-adaptive digital twin framework for multi-task vibration-based assessment. A Transferable Multi-Task Vibration Fusion Network (Trans-MTVFNet) was developed for damage detection, localization, and characterization. The framework utilizes large-scale numerical simulation for pretraining and integrates two domain adaptation strategies to enhance transfer to experimental data. Validated via laboratory-scale tests, results show that pre-training provides a useful initialization, while domain adaptation significantly improves cross-domain generalization for fine-grained tasks. Compared to CNN and transformer benchmarks, Trans-MTVFNet achieves superior accuracy with substantially reduced computational costs. The proposed framework enables efficient and adaptive digital twin-based tunnel damage assessment, providing a promising framework for future vibration-based monitoring of shield tunnels.

1. Introduction

Shield tunnels are critical components of urban transportation infrastructure and are expected to operate safely over long service periods. However, during long-term service periods, tunnels are inevitably subjected to various forms of structural deterioration, such as crack, leakage, voids behind segments, and concrete spalling. These defects may compromise structural integrity and operational safety if not detected and evaluated in a timely manner. As a result, reliable damage detection and condition assessment of in-service shield tunnels have become increasingly important for modern infrastructure management.

Digital twin (DT) technology has emerged as a promising paradigm for structural health monitoring, enabling the integration of physical assets with virtual models to support real-time condition awareness and decision-making [1]. In recent years, data-driven approaches based on deep learning have been widely explored for tunnel damage detection and condition assessment of tunnel structures, providing essential data and models for the development of DTs [2]. Existing studies predominantly focus on image-based recognition of visible tunnel defects, such as surface cracks [3,4] and lining deterioration [5], where rich visual features can be directly extracted. However, many critical tunnel defects

are invisible or weakly observable, including hidden concrete spalling in tunnel segments, fractures, and air- or water-filled voids behind tunnel linings. These types of damage cannot be reliably identified through visual inspection and instead require alternative sensing modalities, such as vibration-based measurements [6,7] or ground penetrating radar [8], for effective detection and evaluation. Building on such sensing techniques, recent studies have increasingly explored deep learning models to analyze experimental and field-acquired data for invisible tunnel damage assessment. CNN-based methods are commonly adopted due to their hierarchical feature extraction capability and relatively low computational overhead, making them suitable for engineering applications with constrained computational resources [9,10]. Moreover, transformer-based architectures have been introduced to capture long-range dependencies and global correlations in vibration responses, achieving improved accuracy in certain damage identification tasks [11]. However, these performance gains are often accompanied by substantially increased computational cost, longer training time, and higher memory consumption. This trade-off between accuracy and computational efficiency poses challenges within DT frameworks that require frequent and efficient model updating. Moreover, most existing methods implicitly assume the availability of sufficient labeled

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experimental data for training and validation. In real in-service tunnel environments, however, damaged samples are inherently scarce compared with undamaged data, as damage events are infrequent and difficult to measure under controlled conditions. While training diagnostic models purely on numerical simulations appears to be a viable workaround, it typically suffers from significant performance degradation when applied to real structures, owing to the inevitable “domain gap” between idealized simulations and complex physical realities. This challenge becomes more pronounced in multi-task damage assessment scenarios, where models are required not only to detect the presence of damage, but also to perform damage localization, severity estimation, or multi-type damage classification simultaneously [5,12]. These practical constraints further limit the robustness and generalizability of existing approaches.

In contrast to experimental or field data, numerical simulations can generate large-scale datasets under controlled damage scenarios, offering a practical means to enrich training data and expose models to diverse damage patterns. By pretraining models on simulation data, transferable damage-related representations can be learned before adapting to real measurements, thereby reducing dependence on extensive labeled experimental data [13]. However, directly transferring models trained on simulation data to real-world conditions is nontrivial due to inevitable domain discrepancies arising from simplified modeling assumptions, sensing noise, and environmental variability [14]. Domain adaptation (DA) techniques have therefore been introduced to bridge this gap by aligning feature representations across domains. Various DA strategies have been proposed, especially in machinery fault diagnosis, including feature-level alignment [15], adversarial DA [16], and class-conditional adaptation [17], which aim to reduce distribution gaps between source and target domains while preserving task-relevant information. However, directly applying these DA techniques to large-scale civil infrastructure, such as shield tunnels, presents unique challenges. Unlike rotating machinery, which operates under high-frequency, periodic excitations within relatively compact components, shield tunnels are massive structures embedded in complex geological environments and subjected to highly non-stationary, low-frequency dynamic train loads. Consequently, their vibration signals are significantly distorted by complex wave attenuation within the concrete lining and unpredictable soil-structure interactions. To address these structural complexities, the proposed two-stage updating scheme is tailored for tunnel environments. In the first stage, pretraining on high-fidelity numerical simulations captures the fundamental physical laws of wave propagation across massive tunnel rings. In the second stage, the dual-strategy DA module acts as a targeted bridge across the reality gap: the global alignment mechanism suppresses macroscopic discrepancies caused by soil boundaries, while the class-conditional alignment isolates the subtle, highly localized signal variations induced by specific structural defects.

To effectively implement this tailored DA scheme within a continuous DT framework, the underlying feature extraction backbone must be capable of resolving these complex spatial-spectral features without imposing prohibitive computational burdens. Fortunately, recent advances in convolutional neural network design have demonstrated that modern ConvNets can achieve performance comparable to transformer-based models [18]. By adopting large-kernel depthwise convolutions, simplified block structures, and training strategies inspired by Transformers, these modern ConvNets enable stronger representation learning and improved global context modeling while preserving the inherent efficiency of convolutional operations. Motivated by these considerations, this study adopts an efficient convolutional network as the backbone and integrates simulation-based transfer learning with DA to support robust model updating. The proposed framework aims to balance accuracy, efficiency, and adaptability within a digital twin-oriented tunnel damage assessment setting.

In summary, this paper proposed a domain-adaptive DT framework for vibration-based multi-task damage detection and assessment of in-

service shield tunnels. The proposed framework integrates simulation-based transfer learning, efficient convolutional modeling, and domain adaptation-driven model updating to enable robust and practical DT deployment. In the context of this study, a true DT is not merely a static numerical simulation or a standalone machine learning model; rather, it requires a physics-based virtual entity, a physical asset, and a dynamic synchronization mechanism between them. Here, the high-fidelity finite element (FE) model serves as the virtual entity, providing a comprehensive baseline of structural mechanics. The proposed DA driven deep learning network (Trans-MTVFNet) acts as the critical computational bridge. Instead of relying on computationally prohibitive real-time FE model updating, the DA module serves as the dynamic synchronization mechanism, continuously aligning the virtual feature space with the real-time physical monitoring data to ensure the condition assessment remains accurate over time.

The main contributions of this paper are summarized as follows:

1. A Transferable Multi-Task Vibration Fusion Network (Trans-MTVFNet) is proposed for joint damage presence identification, ring-level localization, and damage characterization. Built upon a modern and computationally efficient ConvNet backbone, Trans-MTVFNet achieves a favorable balance between representational capacity and efficiency, making it well suited for DT applications.
2. A domain-adaptive DT framework for tunnel damage assessment is developed, which systematically integrates numerical simulation data and experimental measurements within a unified two-stage updating scheme to support robust model transfer and performance improvement in real tunnel environments.
3. A simulation-driven transfer learning and DA strategy is introduced to explicitly enhance task performance during cross-domain transfer. By combining global feature alignment and class-wise DA, the proposed approach improves cross-domain generalization, with noticeable benefits for fine-grained tasks such as ring-level damage identification.
4. Comprehensive experimental validation is conducted using laboratory-scale tests involving different tunnel damage scenarios, including voids and concrete spalling. The results demonstrate the effectiveness, robustness, and computational efficiency of the proposed framework.

2. Methodology

This section details the proposed domain-adaptive DT framework for tunnel damage assessment. It first provides an overview of the two-stage methodology, followed by the specific data processing procedures, the architecture of Trans-MTVFNet, and the domain adaptation strategies utilized for model updating.

2.1. Overview of the method

This paper proposed a two-stage, domain-adaptive DT framework for tunnel damage detection and evaluation, integrating simulation-based pretraining with domain adaptation-driven transfer learning. The overall methodology is illustrated in Fig. 1.

In Stage 1, large-scale simulation data are generated to model diverse tunnel damage scenarios. Raw vibration signals are first synthesized under different damage conditions and then transformed into time-frequency representations. Based on these simulated signals, the proposed network is pretrained to learn generic and transferable damage-related representations. This stage focuses on general feature learning, enabling the model to capture fundamental structural response patterns associated with damage while avoiding overfitting to specific experimental conditions.

In Stage 2, the pretrained model is adapted to laboratory data through a supervised domain adaptation setting. Specifically, labeled experimental vibration signals collected from physical tunnel models are

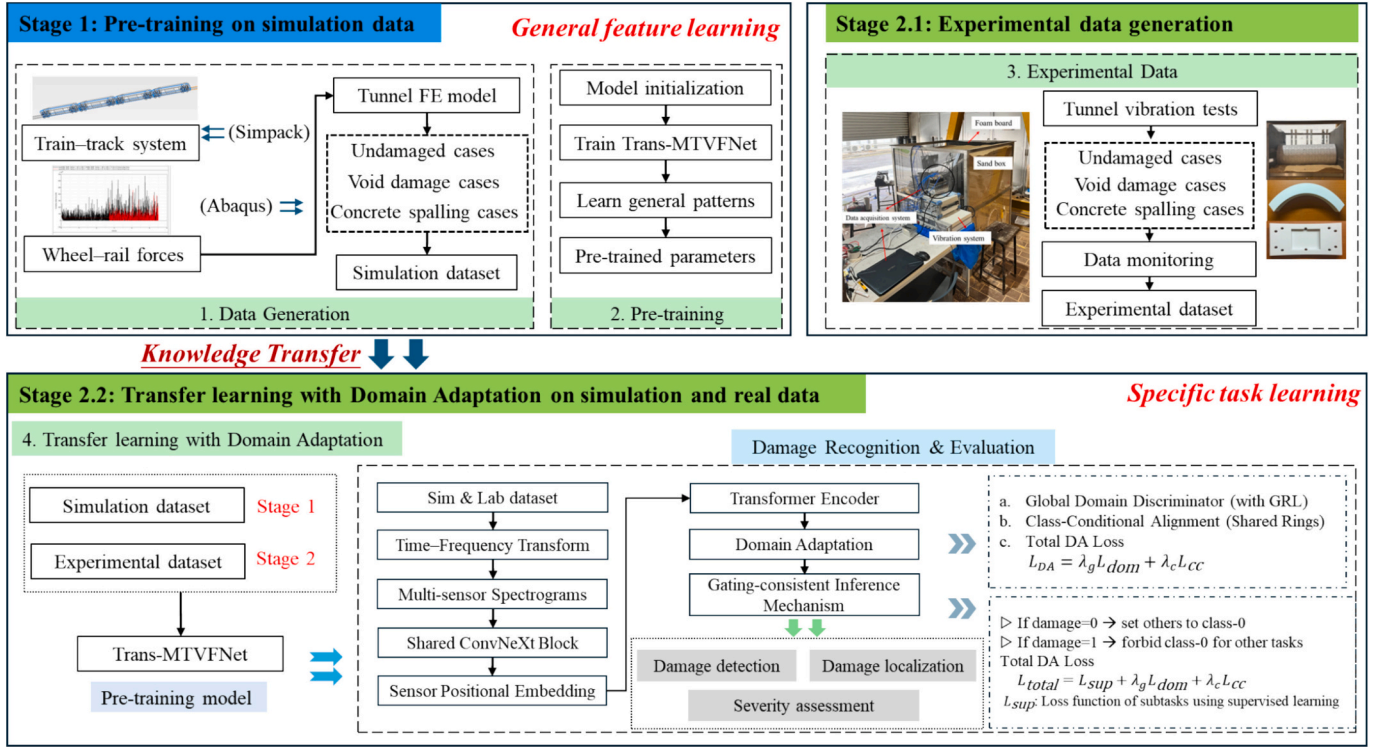


Fig. 1. Framework of the proposed method.

used together with labeled simulated data. This supervised transfer learning process aims to mitigate domain discrepancies arising from sensing noise, material properties, and testing environments, while leveraging target-domain labels for accurate task refinement. DA is incorporated during training to align feature distributions across simulation and laboratory domains, allowing effective knowledge transfer while preserving task-relevant information. This stage emphasizes task-specific learning, refining the model for accurate damage recognition under real-world conditions.

The proposed framework supports multiple damage-related tasks, including damage detection, damage localization, and severity assessment. By combining simulation-driven representation learning with domain-adaptive transfer, the DT evolves from a physics-informed simulation model into a data-driven and deployment-ready system for

tunnel damage evaluation.

2.2. Data processing

To transform raw tunnel vibration measurements into a network-ready representation, the acceleration time series collected from multiple sensors are processed through a series of signal conditioning and feature extraction procedures. The goal is to obtain stable, informative, and spatially correlated spectrogram features that reflect the tunnel structural dynamics under train loads. Fig. 2 summarizes the complete data processing workflow. The following data processing steps are applied to extract and standardize the raw acceleration signals:

Step 1 Raw Acceleration Signals: The tunnel vibration dataset consists of multi-sensor acceleration time series recorded during train

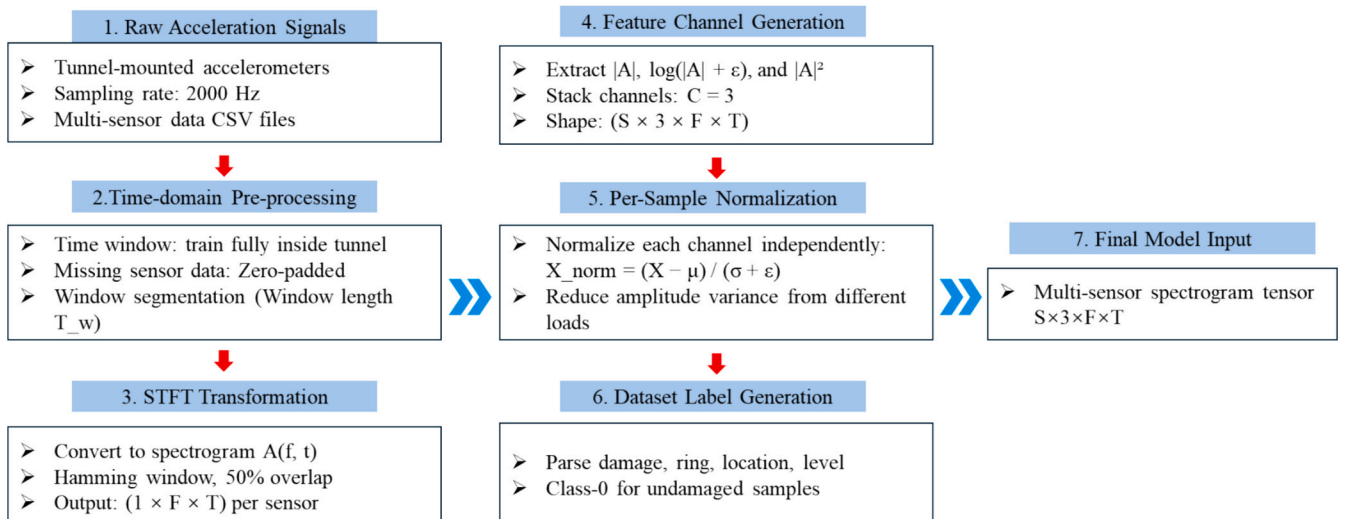


Fig. 2. Data processing workflow.

passage. All samples are collected from multiple accelerometers installed on the lining.

Step 2 Time-domain Pre-processing: Minimal metadata-based cleaning is performed before frequency analysis: Automatic detection of Time, Sensor, and Acceleration fields in data files; Cropping of signals to a desired time range using *time_min* / *time_max*.

Step 3 STFT Transformation: The Short-Time Fourier Transform (STFT) is applied to each sensor signal, converting it into a complex spectrogram of size $(F \times T)$ per sensor. This conversion uses a Hann window with an overlap, where the number of frequency bins (F) is determined by n_{fft} .

Step 4 Feature Channel Generation: Following the STFT operation, each acceleration signal is transformed into a spectrogram $A(f, t)$, where f and t denote the frequency bin and time frame, respectively. To fully leverage both amplitude and spectral-energy information embedded in $A(f, t)$, three complementary representations are derived:

$$\text{Magnitude} : M(f, t) = |A(f, t)|$$

$$\text{Log-magnitude} : L(f, t) = \log(|A(f, t)| + \epsilon) \quad (1)$$

$$\text{Power spectral density} : P(f, t) = |A(f, t)|^2$$

Here, ϵ is a small positive constant used for numerical stability. By stacking the three maps channel-wise, each sensor output becomes a compact 3-channel feature tensor:

$$X \in \mathbb{R}^{S \times C \times F \times T} \quad (C = 3) \quad (2)$$

where S is the sensor number, F is the frequency resolution, and T is the number of time frames. This “RGB-like” construction provides complementary physics, enabling the network to jointly exploit amplitude, contrast enhancement, and energy-distribution cues without additional hand-crafted filtering rules.

Step 5 Per-Sample Normalization: To eliminate amplitude variations caused by different train loads or sensor gains, each channel of each sensor is normalized independently:

$$X_{\text{norm}} = \frac{X - \mu}{\sigma + \epsilon_1} \quad (3)$$

where μ and σ denote the mean and standard deviation of channel c from sensor s , and ϵ_1 is a stabilizing constant.

Step 6 Dataset label generation: The ground-truth labels of each sample are parsed directly from the file name, which encodes the tunnel condition simulated or measured in the experiment. Four attributes are extracted: (1) damage presence, (2) ring index, (3) damage location, and (4) severity level. Undamaged samples are assigned label 0 for all attributes, while damaged samples keep their corresponding metadata.

Step 7 Final Model Input: After STFT channel construction and normalization, the data are organized into a unified tensor format:

$$X \in \mathbb{R}^{B \times S \times 3 \times F \times T} \quad (4)$$

where B is the batch size, S is the number of sensors, and the three channels represent magnitude, log-magnitude, and PSD. This standardized representation ensures that both simulated and experimental data share consistent dimensions and can be directly fed into the proposed network for multi-task tunnel damage recognition.

2.3. Trans-MTVFNet

To fully exploit the rich dynamic information captured by multi-sensor vibration measurements, a Transferable Multi-Task Vibration Fusion Network (Trans-MTVFNet) is developed (see Fig. 3). The network aims to learn generalized vibration representations from synthetic simulations and adapt them to experimental conditions, enabling reliable tunnel damage detection with limited real data. First, it employs a

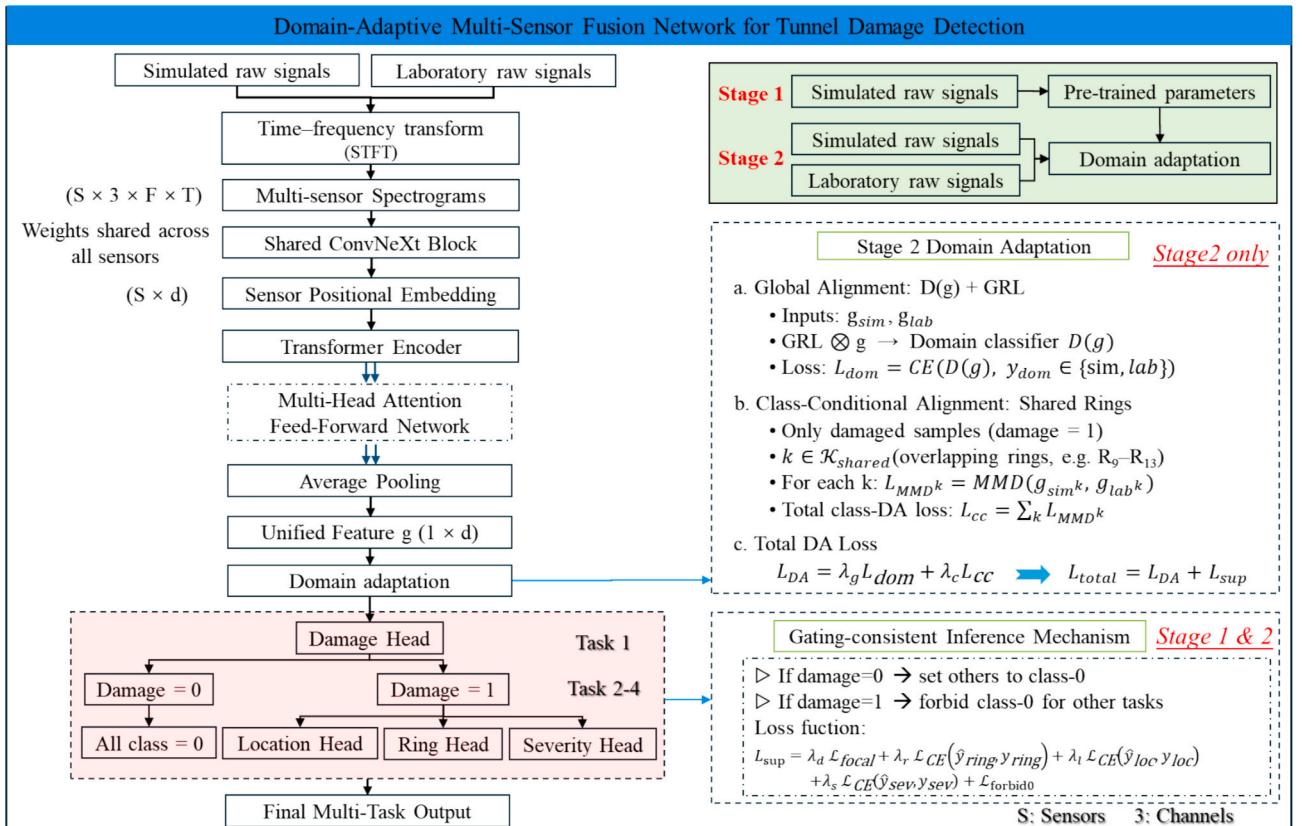


Fig. 3. Two-stage domain-adaptive framework for tunnel damage detection.

modern ConvNet backbone to extract local spectral representations from multi-sensor time-frequency spectrograms efficiently. Second, instead of applying self-attention to raw signals, it integrates a lightweight Transformer Encoder specifically on the latent features to model the cross-sensor spatial correlations driven by structural load propagation. Finally, Trans-MTVFNet incorporates a physics-informed, task-coupled gating strategy. This mechanism ensures that fine-grained tasks (localization and severity assessment) are conditionally activated only when a positive damage state is detected, thereby enforcing semantic consistency and avoiding logical contradictions in multi-task predictions. Therefore, through a joint learning strategy, the model simultaneously performs damage identification, damage localization, and severity estimation, thereby providing a comprehensive assessment of tunnel structural condition.

As shown in Fig. 3, Stage 1 corresponds to the simulation-based training phase used for model pretraining. To effectively extract general damage-related features from large-scale simulated data, a lightweight yet expressive network architecture is adopted. Specifically, the depthwise-pointwise decomposition in ConvNeXt blocks enhances representational power while maintaining computational efficiency. The feature transformation in the ConvNet block can be expressed as:

$$\mathbf{y} = \mathbf{x} + \text{PWConv}(\text{DWConv}(\mathbf{x})) \quad (5)$$

where $\mathbf{x}$ and $\mathbf{y}$ are the input and output of ConvNet blocks, DWConv captures long-range spatial-spectral structures, and PWConv mixes cross-channel interactions efficiently.

Each sample is expressed as a time-frequency tensor $\mathbf{X}$. A shared ConvNet encoder f_θ extracts spectral representations from individual sensors:

$$\mathbf{z}_s = f_{\theta_sh}(\mathbf{X}_s), \mathbf{z}_s \in \mathbb{R}^d \quad (6)$$

where $\mathbf{z}_s$ are latent features and θ_{sh} are the shared weights. These latent features $\mathbf{Z}$ are then enhanced by a self-attention fusion module, which models cross-sensor spatial correlations caused by loading propagation in the lining:

$$\mathbf{Z} = \text{SoftAttn}([\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_S]) \in \mathbb{R}^{S \times d} \quad (7)$$

Global vibration descriptors $\mathbf{g}$ are obtained by averaging:

$$\mathbf{g} = \frac{1}{S} \sum_{s=1}^S \mathbf{z}_s \in \mathbb{R}^d \quad (8)$$

This fusion procedure ensures that the final representation encodes both local sensor responses and global structural coupling effects. Based on this rich representation, the subsequent learning framework is designed to align with the hierarchical nature of structural damage assessment (i.e., detection, localization, and quantification). By formulating this as a multi-task learning problem, the network optimizes a shared feature representation across all tasks, which provides mutual regularization and prevents overfitting to a single objective. To achieve this, three task-specific prediction heads are defined on the shared fused vector $\mathbf{g}$:

$$\begin{aligned} \hat{\mathbf{y}}_{\text{damage}} &= \sigma(\mathbf{W}_d \mathbf{g} + \mathbf{b}_d) \\ \hat{\mathbf{y}}_{\text{ring}} &= \text{Softmax}(\mathbf{W}_r \mathbf{g} + \mathbf{b}_r) \\ \hat{\mathbf{y}}_{\text{loc}} &= \text{Softmax}(\mathbf{W}_l \mathbf{g} + \mathbf{b}_l) \\ \hat{\mathbf{y}}_{\text{sev}} &= \text{Softmax}(\mathbf{W}_s \mathbf{g} + \mathbf{b}_s) \end{aligned} \quad (9)$$

where $\hat{\mathbf{y}}_{\text{damage}} \in [0, 1]$ is probability of damage presence, $\hat{\mathbf{y}}_{\text{ring}}$ stands for ring-level localization, $\hat{\mathbf{y}}_{\text{loc}}$ stands for damage location within the corresponding ring, $\hat{\mathbf{y}}_{\text{sev}}$ is severity classification, $\mathbf{W}_{(\cdot)}$ and $\mathbf{b}_{(\cdot)}$ are trainable parameters in each task head, and $\sigma(\cdot)$ is sigmoid activation function.

Since localization and severity predictions are only meaningful when damage exists, a task-coupled gating strategy is applied during inference

to ensure semantic consistency. The proposed gated inference mechanism enforces physical logic: the localization and severity assessment heads are activated conditionally, only when the initial damage detection head outputs a positive (damaged) state. This prevents the model from making nonsensical predictions (e.g., predicting a severe damage level for a healthy tunnel ring) and computationally streamlines the inference process:

$$\tilde{\mathbf{y}}_{\text{damage}} = 1 \left(\hat{\mathbf{y}}_{\text{damage}} \geq \tau \right) \quad (10)$$

where $\tau \in (0, 1)$ is the damage decision threshold and $\tilde{\mathbf{y}}_{\text{damage}} \in \{0, 1\}$ is the binarized damage prediction. Then, the final predictions of localization and severity tasks are gated as:

$$\begin{aligned} \tilde{\mathbf{y}}_{\text{ring}} &= \begin{cases} \text{argmax}_k \hat{\mathbf{y}}_{\text{ring},k}, & \tilde{\mathbf{y}}_{\text{damage}} = 1 \\ 0, & \tilde{\mathbf{y}}_{\text{damage}} = 0 \end{cases} \\ \tilde{\mathbf{y}}_{\text{loc}} &= \begin{cases} \text{argmax}_k \hat{\mathbf{y}}_{\text{loc},k}, & \tilde{\mathbf{y}}_{\text{damage}} = 1 \\ 0, & \tilde{\mathbf{y}}_{\text{damage}} = 0 \end{cases} \\ \tilde{\mathbf{y}}_{\text{sev}} &= \begin{cases} \text{argmax}_k \hat{\mathbf{y}}_{\text{sev},k}, & \tilde{\mathbf{y}}_{\text{damage}} = 1 \\ 0, & \tilde{\mathbf{y}}_{\text{damage}} = 0 \end{cases} \end{aligned} \quad (11)$$

where class 0 explicitly denotes “no-damage” state for both localization and severity.

In tunnel condition monitoring, most samples are healthy (no-damage), while damaged cases are very limited, especially severe ones. To address this, focal loss introduces a prediction-dependent modulation factor:

$$\mathcal{L}_{\text{focal}} = -\alpha(1-p)^\gamma \log(p) - (1-\alpha)p^\gamma(1-y)\log(1-p) \quad (12)$$

where $p = \hat{\mathbf{y}}_{\text{damage}}$ is predicted positive probability, $y \in \{0, 1\}$ stands for true damage label, $\alpha \in [0, 1]$ is class-balancing weight, and $\gamma \geq 0$ is focusing parameter to down-weight well-classified samples.

To prevent meaningless predictions for damaged samples (i.e., predicting class-0 for positive damage), a zero-class suppression regularizer is added:

$$\mathcal{L}_{\text{forbid0}} = \beta(\text{softplus}(\mathbf{z}_{l,0}) + \text{softplus}(\mathbf{z}_{s,0})) \quad (13)$$

where $\mathbf{z}_{l,0}$ and $\mathbf{z}_{s,0}$ are logits of class-0 in localization and severity heads, β is penalty weight, and $\text{softplus}(x) = \log(1 + e^x)$ ensures numerical stability.

The overall objective is a weighted combination of focal loss for damage classification and cross-entropy for the other two tasks:

$$\begin{aligned} \mathcal{L}_{\text{sup}} &= \lambda_d \mathcal{L}_{\text{focal}} + \lambda_r \mathcal{L}_{\text{CE}}(\hat{\mathbf{y}}_{\text{ring}}, \mathbf{y}_{\text{ring}}) + \lambda_l \mathcal{L}_{\text{CE}}(\hat{\mathbf{y}}_{\text{loc}}, \mathbf{y}_{\text{loc}}) + \lambda_s \mathcal{L}_{\text{CE}}(\hat{\mathbf{y}}_{\text{sev}}, \mathbf{y}_{\text{sev}}) \\ &\quad + \mathcal{L}_{\text{forbid0}} \end{aligned} \quad (14)$$

where λ_d , λ_r , λ_l , and λ_s are balancing weights among the four tasks, $\mathcal{L}_{\text{CE}}$ is standard cross-entropy loss, and $\mathbf{y}_{\text{ring}}$, $\mathbf{y}_{\text{loc}}$ and $\mathbf{y}_{\text{sev}}$ are ground-truth labels for localization and severity. This gating mechanism enforces logical consistency among the three tasks, avoids error propagation from false positives, and improves robustness when real measurement noise exists.

Using a common backbone promotes joint learning of damage-related representations, where global anomaly features support precise localization and severity estimation, and detailed local cues improve overall detection reliability. Stage 1 serves as the simulation-based training phase, where the model is pretrained. The learned parameters are subsequently transferred to Stage 2.

2.4. Domain-adaptive DT modeling

Stage 2 adapts the pretrained DT model from simulation to laboratory data by addressing domain discrepancies between the two sources (see Fig. 3). While Stage 1 learns generic damage-related representations from simulated signals, differences in sensing noise, material properties, and experimental conditions introduce domain shifts that hinder direct deployment. Therefore, the pretrained parameters are used as initialization, and supervised DA is performed in Stage 2 using both simulated and laboratory data. During this model updating phase, the entire Trans-MTVFNet architecture, including the shared ConvNeXt backbone, the Transformer encoder, and all task-specific heads, undergoes full fine-tuning. This full fine-tuning strategy is chosen because the domain gap between idealized numerical simulations and physical tunnel measurements is substantial. By continuously updating all network parameters, the ConvNeXt backbone can optimally adjust its low-level spectral feature extraction capabilities to accommodate the specific environmental noise and boundary conditions of the target domain, while the DA losses concurrently regularize the feature alignment to prevent catastrophic forgetting of the pre-trained knowledge. Under a supervised setting, the target-domain (laboratory) samples used for model updating are fully labeled, allowing the network to compute task-specific supervised losses alongside the domain alignment constraints. Adaptation is applied at the unified feature level g , enabling the model to preserve transferable structural knowledge while aligning domain-specific representations. Two complementary DA strategies are employed to achieve robust and task-consistent cross-domain transfer.

The first DA strategy focuses on reducing the global distribution discrepancy between simulated and laboratory features. Given unified representations g_{sim} and g_{lab} , a domain discriminator $D(\cdot)$ is introduced to distinguish the feature origin.

A gradient reversal layer (GRL) is applied before the domain discriminator, encouraging the backbone network to learn domain-invariant representations through adversarial optimization. The discriminator is trained to classify the domain label $y_{\text{dom}} \in \{\text{sim}, \text{lab}\}$, while the feature extractor is optimized to confuse the discriminator. The corresponding domain loss is defined as:

$$L_{\text{dom}} = \text{CE}(D(g), y_{\text{dom}}) \quad (15)$$

where CE denotes the cross-entropy loss. This global alignment mechanism primarily targets coarse distribution shifts, enabling the DT to capture domain-agnostic anomaly patterns while suppressing domain-specific artifacts introduced by different data acquisition processes.

While global alignment promotes overall feature consistency, it may overlook fine-grained, task-relevant discrepancies associated with damage characteristics [19]. To address this issue, a second DA strategy is introduced based on class-conditional alignment. This strategy is applied only to damaged samples (damage = 1) and focuses on structural regions that are shared across domains. Specifically, alignment is performed on overlapping ring indices $k \in \mathcal{R}_{\text{shared}}$ between simulation and laboratory data. For each shared ring k , a Maximum Mean Discrepancy (MMD) loss is computed between the corresponding features:

$$L_{\text{MMD}}^k = \text{MMD}(g_{\text{sim}}^k, g_{\text{lab}}^k) \quad (16)$$

The total class-conditional DA loss is then defined as:

$$L_{\text{cc}} = \sum_{k \in \mathcal{R}_{\text{shared}}} L_{\text{MMD}}^k \quad (17)$$

By restricting alignment to damaged samples and shared structural regions, this strategy preserves discriminative damage cues while avoiding negative transfer from irrelevant or non-overlapping regions.

The total DA loss in Stage 2 combines both global and class-conditional alignment terms:

$$L_{\text{DA}} = \lambda_g L_{\text{dom}} + \lambda_c L_{\text{cc}} \quad (18)$$

where λ_g and λ_c balance the contributions of the two adaptation strategies. The final training objective is given by:

$$L_{\text{total}} = L_{\text{DA}} + L_{\text{sup}} \quad (19)$$

where L_{sup} denotes the supervised multi-task loss defined in Section 2.3.

Through the joint application of global and class-conditional DA, the proposed framework enables robust DT transfer from simulation to laboratory environments while maintaining task consistency across damage detection, localization, and severity estimation.

To comprehensively evaluate the performance of the proposed domain-adaptive DT model, weighted average precision, weighted average recall, and weighted average F1-score are adopted as the main evaluation metrics. These metrics are particularly suitable for multi-class and imbalanced damage detection tasks, as they account for both class-wise performance and sample distribution.

Assume a multi-class classification problem with C classes. For each class c , precision, recall, and F1-score are defined as:

$$\begin{aligned} \text{Precision}_c &= \frac{TP_c}{TP_c + FP_c} \\ \text{Recall}_c &= \frac{TP_c}{TP_c + FN_c} \\ \text{F1}_c &= \frac{2 \cdot \text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c} \end{aligned} \quad (20)$$

where TP_c , FP_c , and FN_c denote the number of true positives, false positives, and false negatives for class c , respectively.

Let N_c denote the number of samples belonging to class c , and $N = \sum_{c=1}^C N_c$ be the total number of samples. The weighted average precision, recall, and F1-score are computed as:

$$\begin{aligned} \text{Precision}_{\text{weighted}} &= \sum_{c=1}^C \frac{N_c}{N} \cdot \text{Precision}_c \\ \text{Recall}_{\text{weighted}} &= \sum_{c=1}^C \frac{N_c}{N} \cdot \text{Recall}_c \\ \text{F1}_{\text{weighted}} &= \sum_{c=1}^C \frac{N_c}{N} \cdot \text{F1}_c \end{aligned} \quad (21)$$

These weighted metrics account for class imbalance by assigning higher importance to classes with more samples, providing a more representative evaluation of overall model performance in realistic damage detection scenarios.

To apply the proposed framework in practice, its operation in a real-world tunnel monitoring scenario follows a continuous lifecycle. Initially, during the offline construction phase, a high-fidelity numerical model of the target tunnel section is developed to generate large-scale simulated vibration data under various train loads and damage scenarios, which are then used to pretrain the Trans-MTVFNet. Subsequently, for online deployment and initial alignment, this pretrained model is deployed to the physical tunnel's edge-computing or cloud servers. At this point, baseline vibration data collected from physical sensors are utilized to perform the first round of DA, calibrating the model to the specific environmental noise and boundary conditions of the actual tunnel. Finally, during the continuous monitoring phase, the DT ingests real-time sensor data to execute multi-task damage assessments. As the tunnel ages and its structural conditions naturally drift over time, the domain adaptation module is periodically triggered using recent batches of monitoring data to incrementally update the network parameters. Ultimately, this dynamic updating mechanism ensures that

the DT remains faithfully synchronized with its physical counterpart throughout its entire service life.

3. Case study

To validate the effectiveness and robustness of the proposed domain-adaptive DT framework, a comprehensive case study is conducted. This section introduces the laboratory-scale test setup and the corresponding numerical modeling used for data generation, followed by detailed evaluations under two common shield tunnel defect scenarios: voids behind the segment and concrete spalling.

3.1. Test introduction and data collection

The engineering site selected for this study is located along Singapore's Cross-Island Line, and this study focuses on detecting defects such as voids and concrete spalling in MRT tunnels. The tunnel is a single-track structure composed of circular segment linings with staggered joints. Its key specifications are: outer diameter of 12 m, lining thickness of 0.45 m, ring width of 2 m, and tunnel depth of 35 m. The lining is made of C50-grade concrete with an elastic modulus of 35 GPa [20]. Due to the limited availability and high cost of obtaining comprehensive field monitoring data, this study adopts a laboratory-scale testing approach to detect the tunnel defects.

3.1.1. Lab test setup

Figs. 4 and 5 illustrate the Schematic layout of the tunnel test and test apparatus, which is composed of four main components: a model box, an assembled shield tunnel, a vibration system and a data acquisition system. The tunnel model was simulated using 3D printing technology (Acrylonitrile Butadiene Styrene) with a diameter of 35 cm and a thickness of 1.29 cm, with an elastic modulus of 2.15 GPa. In comparison, the actual tunnel has a diameter of 12.2 m, a lining thickness of 0.45 m, and a lining elastic modulus of 35 GPa. Based on the Buckingham π theorem, the primary scale factors were established. The similarity ratios for the model are determined as follows: geometric scale $S_L = 35$, elastic modulus $S_E = 16.3$, and density $S_\rho = 1$. Based on this modulus scale factor and the geometric scaling, the remaining scale factors employed in the model test are derived through similitude analysis, as summarized in Table 1. While the $S_\rho = 1$ assumption implies that gravity-induced static stresses are distorted, this compromise is accepted in vibration-based assessments where the high-frequency dynamic response is primarily dictated by the distribution of mass and bending stiffness.

The tunnel structure and track slab were created using 3D printing technology. A complete lining ring consists of ten segments: a sealing-roof segment, two adjacent segments, and seven standard segments. Although bolts are typically used to assemble lining segments, this experiment simplified the connection by using steel wires applied from the exterior to facilitate assembly. Although the staggered segment joints were simplified using external steel wire tensioning instead of internal bolts, the tunnel lining remains within the linear-elastic regime under the micro-amplitude dynamic loads induced by train operations. Under these serviceability conditions, the joints remain closed, ensuring that the global stiffness matrix and damping characteristics of the laboratory model faithfully represent the prototypical structural dynamics. Geometric details of the shield tunnel (i.e., lining segments, bolt locations, and the track slab) are found in [20]. The lab test simulates two damage conditions, including voids behind the lining and segment concrete spalling. Table 2 and Fig. 6 show geometric and dimensions of the tunnel segments, damaged segments, and voids. The soil surrounding the tunnel is divided into four zones, corresponding to four locations. At each location, voids of four different depths (20 mm, 30 mm, 40 mm, and 50 mm), representing four damage levels, are installed. Note that in Fig. 6(e), only one void depth (i.e., one damage level) is shown for each location for illustration purposes. As the tunnel structure ages, environmental factors progressively weaken the concrete. This deterioration reduces its mechanical strength, leading to surface spalling and a decrease in segment stiffness. To simulate this condition, tunnel lining segments were prepared with controlled spalling defects, each consisting of a 3×3 cm area and depths of 5 mm.

To manage experimental complexity, the moving train load was simplified in the laboratory using a stationary vibration system. This simplification is grounded in the observation that when a train is fully situated within a specific monitored tunnel section, the primary diagnostic information is embedded in the stable time-frequency patterns of the structural response. To evaluate the waveform, amplitude, and frequency characteristics of vibration loads during train operation, Ding et al. [21] employed a train-track-tunnel-foundation dynamic coupling model to obtain the time-history curve of the wheel-rail contact force under vehicle loading. To quantify the numerical values of train-induced vibration loads and frequencies, this study adopted the train load calculation model proposed by Lei et al. [22] to determine their respective ranges. The vibration load and frequency are expressed:

$$F_m = \frac{q_p l_m}{S_E S_L} \quad (22)$$

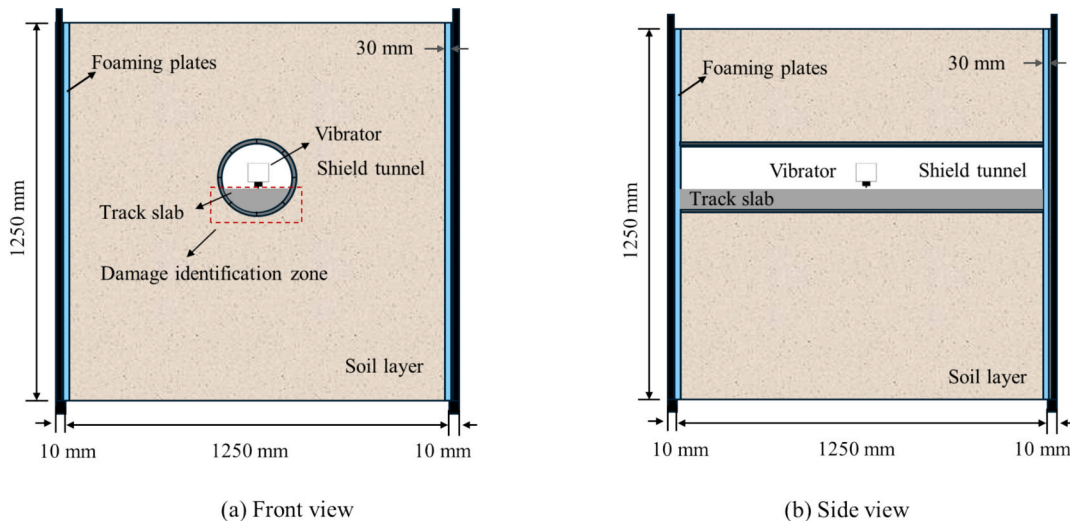


Fig. 4. Schematic layout of the tunnel test: (a) Front view and (b) Side view.

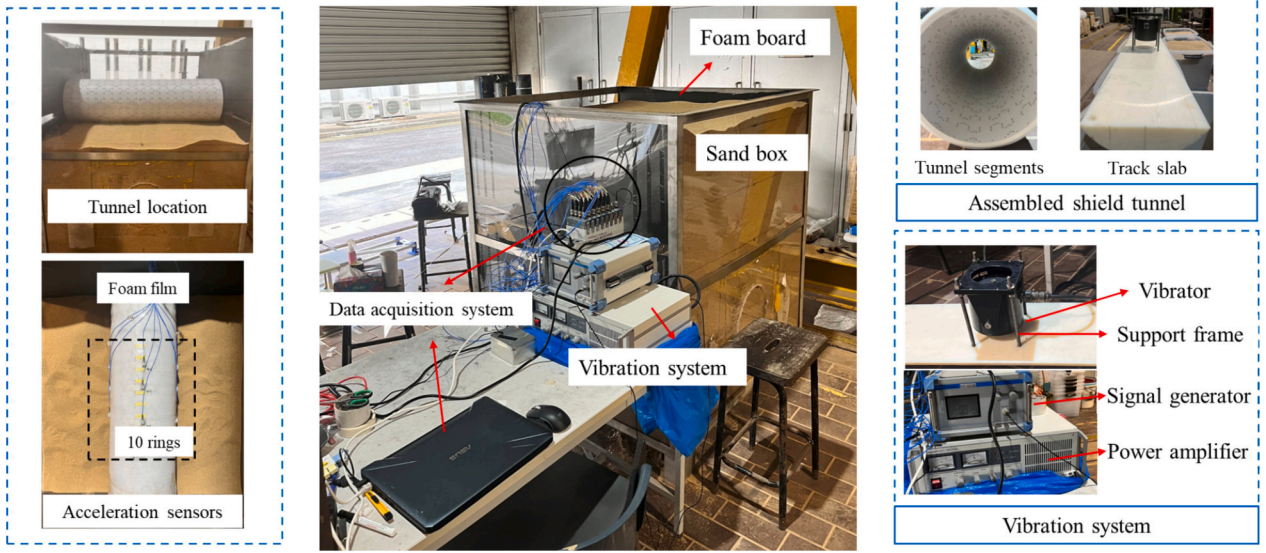


Fig. 5. Test apparatus.

Table 1
Scale factors of the tunnel testbed.

Physical quantities		Similarity ratio
Basic quantities	Length S_L	35
	Elastic modulus S_E	16.3
	Density S_ρ	1
	Stress $S_\sigma = S_E$	16.3
Derived quantities	Displacement $S_u = S_L$	35
	Bending stiffness $S_{EI} = S_E \times S_L^4$	24.5×10^6

Table 2
Key physical parameters of the tunnel system.

Components	Parameters	Value
Sand box	Length×width×height (m)	$1.2 \times 1.2 \times 1.2$
	Tunnel	
Tunnel	External diameter (m)	0.35
	Length (m)	1.2
	Tunnel depth (m)	0.4
	Ring width (mm)	58
	Lining thickness (mm)	13
Track plate	Length (m)	1.2
	Diameter (m)	0.35
Foam board	Thickness (mm)	10

$$f_m = \frac{S_L}{T_p} \sqrt{\frac{S_p}{S_E}} \quad (23)$$

In the equations, q_p represents the uniformly distributed load on the track, l_m denotes the influence range of the vibration load in the model, which is set to 1.25 m in this study, and T_p is the duration of the train load. In the experiment, the train-induced load was simulated using the vibration system. The vibration system consists of a signal generator, a power amplifier, and a vibrator. During the experiment, the force signal is first input into the signal generator, and the vibration force is adjusted using feedback from force sensors to regulate the power amplifier. This process generates the required vibration frequency and force, which are then transmitted to the tunnel via a track plate, simulating train-induced vibrations. It is important to note that to ensure the stable transmission of vibration force, a rigid connection is established between the vibrator and the track slab, as is shown in Fig. 6 (d). For the Singapore EWL trains, the approximate train length is 140 m, the total unladen weight is about 228 t, and the typical operating speed is around 80 km/h. Under a

maximum passenger density of six person/m², and assuming an average passenger mass of 60 kg, the fully loaded train weight can be estimated accordingly. The calculated vibration force F_m is approximately 48 N, and the vibration frequency f_m is about 2 Hz.

The experiment consists of three parts: the undamaged tunnel case, the void-behind-lining case, and the concrete spalling case. During the experiment, different train loading conditions and operational speeds were simulated in the experiment by varying the vibration force and frequency. Because the number of passengers on a train varies randomly, the overall train load also fluctuates. Therefore, the vibration forces were set to 40 N and 50 N, and the vibration frequency was chosen as 2 Hz. Table 3 shows test conditions. Fig. 7 shows the schematic diagram of the tunnel and monitoring plan. To minimize boundary effects, the central ten rings were designated as the tunnel damage monitoring zone, and to reduce the experimental workload, damage was introduced in only five of these rings. As shown in Fig. 7, the three-axis acceleration sensors are installed in the tunnel structure with a sampling frequency of 2000 Hz. The vertical acceleration response was used to verify the effectiveness of the proposed method.

3.1.2. FE modeling

To investigate how different tunnel lining damage scenarios respond under realistic operational loads, this study adopts a two-stage numerical modeling framework. First, the wheel-rail interaction forces generated by a running train are obtained from a detailed train-track dynamic model. Then, these dynamic loads are applied to a three-dimensional tunnel-track finite element model to evaluate the structural response under both healthy and damaged conditions. Based on these simulations, a comprehensive dataset is constructed for subsequent analysis of damage signatures and model development. The overall workflow of this study is illustrated in Fig. 8, integrating Simpack for dynamic load acquisition and Abaqus for tunnel structural analysis.

Fig. 9 presents a schematic of the FE model, which has dimensions of 80 m (width) × 80 m (length) × 70 m (height). The model details can be found in [23]. The shield tunnel features a single-track design with circular segment linings that include staggered joints. This FE model is constructed with 169,854 nodes and 126,953 elements. To enhance computational efficiency, a multiscale hybrid modeling approach is employed for the simulation of segmental linings. Specifically, the lining rings situated at the two sides of the existing tunnel are approximated as a continuous cylindrical ring, assuming a linear elastic behavior. The track is simulated using beam elements, and the track is connected by

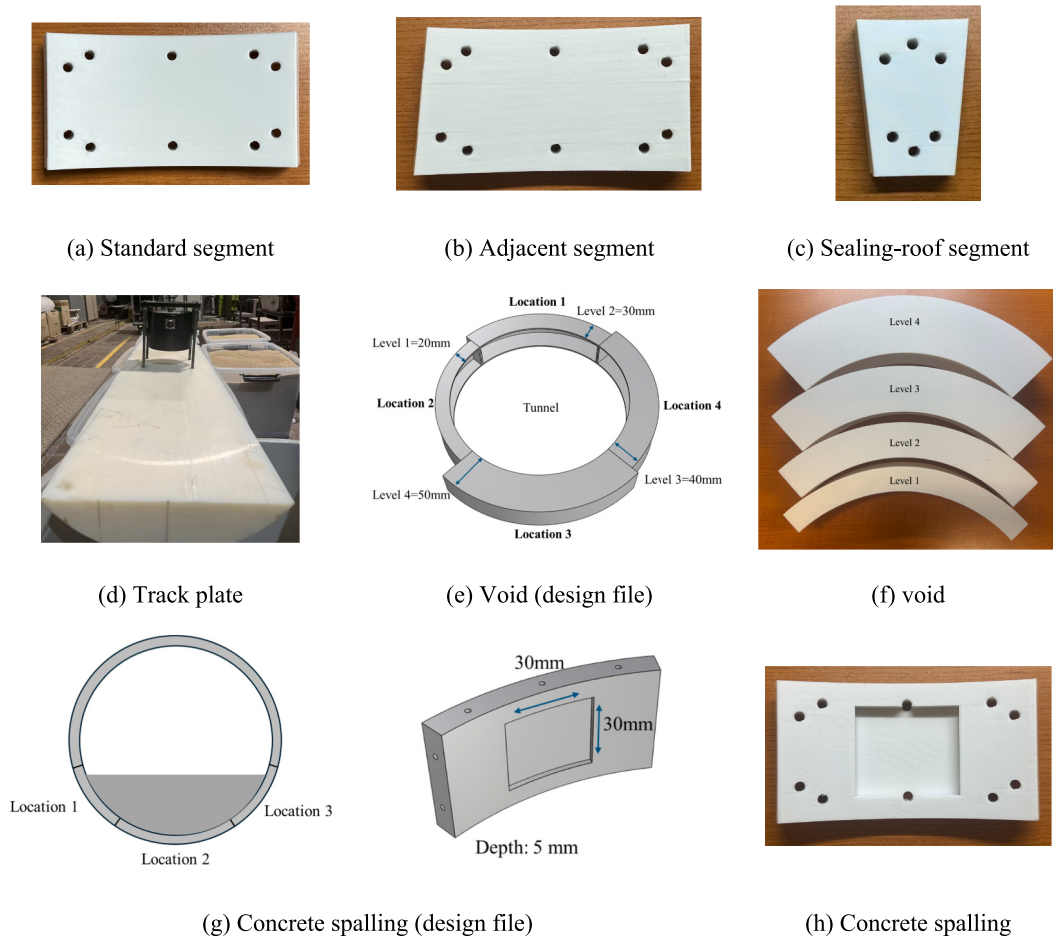


Fig. 6. Tunnel segments and damage setup.

Table 3

Test conditions.

Case	Ring number (1–10)	Damage location	Damage level
Undamaged	–	–	–
Void	4	1/2/3/4	1/2/3/4
	5	1/2/3/4	1/2/3/4
	6	1/2/3/4	1/2/3/4
	7	1/2/3/4	1/2/3/4
	8	1/2/3/4	1/2/3/4
Concrete spalling	4	1/2/3	1
	5	1/2/3	1
	6	1/2/3	1
	7	1/2/3	1
	8	1/2/3	1

connectors. The numerical model adopts a multiscale hybrid boundary to approximate energy dissipation into the surrounding soil. These structural simplifications, along with differences between the simulated infinite soil boundaries and the physical sandbox boundaries, introduce systematic dynamic discrepancies between the numerical and laboratory models. However, it is precisely these unavoidable physical modeling gaps that necessitate the proposed DA module, which is designed to robustly align the target features and mitigate the impact of such experimental and numerical idealizations. Tables 4 and 5 shows parameters of modeling components. For the displacement boundary conditions, normal motions on both sides of the 3D model are restricted. The upper boundary of the model is unconstrained, while the lower boundary is fixed and restricted from movement in all three dimensions. To minimize boundary condition effects, infinite elements are applied to the boundaries of the model.

A four-carriage train-track subsystem has been successfully established in Simpack (See Fig. 10), with carriages connected using force elements and parameters derived from the literature [24,25]. To enhance fidelity while maintaining computational efficiency, Linear Flexible Tracks have been imported from Abaqus, and practical unevenness has been incorporated into the track definition. The vertical wheel-rail interaction forces are collected and utilized as input for the dynamic analysis (Dynamic, Implicit) of the tunnel in Abaqus via the Dload subroutine. Fig. 11 shows the results of interaction forces at 2.7 s. Two rails have been constructed atop the tunnel concrete plate in Abaqus, enabling the application of time-dependent wheel-rail vertical forces in a spatio-temporal manner.

The detailed simulation procedures for data generation are summarized as follows:

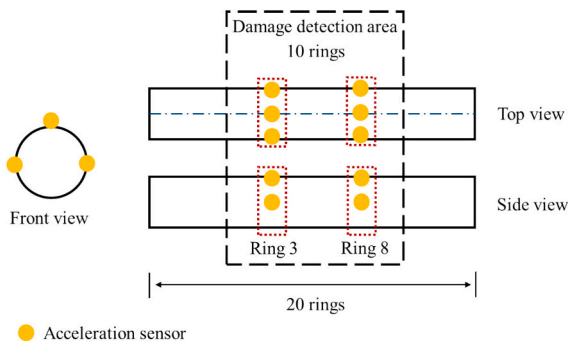


Fig. 7. Schematic diagram of the tunnel and monitoring plan.

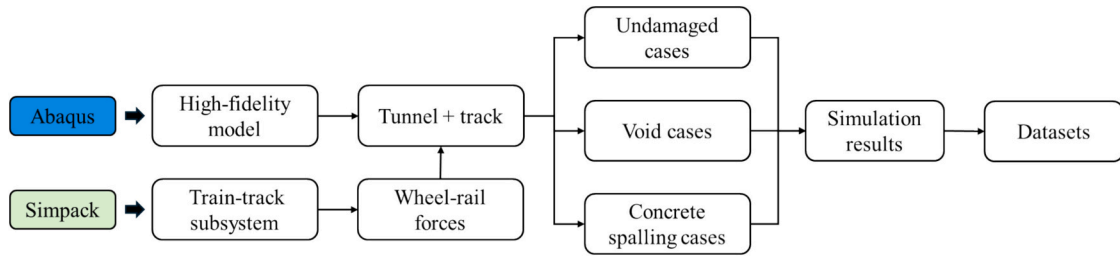


Fig. 8. Workflow of data generation in simulation models.

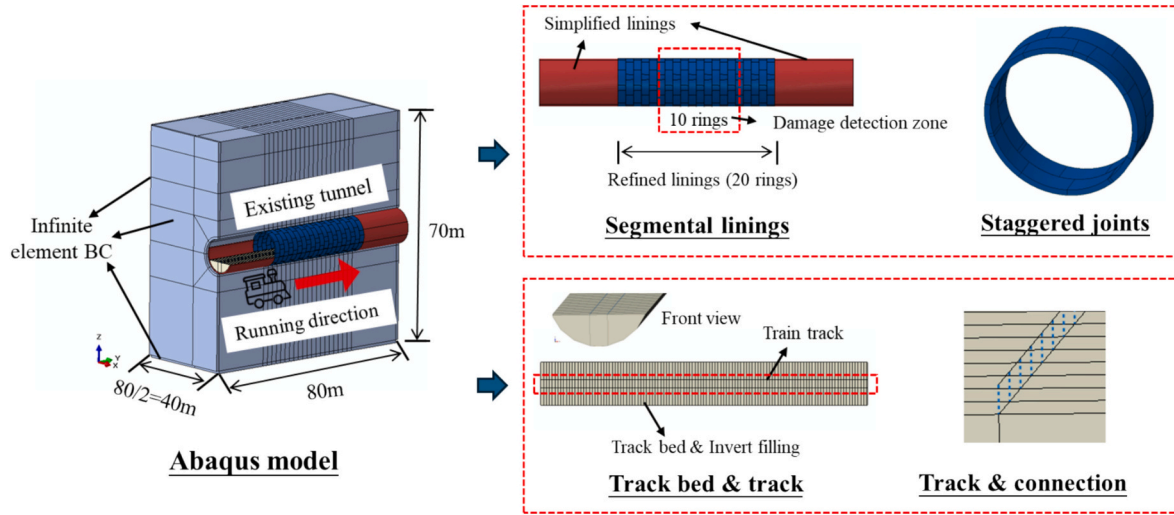


Fig. 9. FE model in Abaqus.

Table 4
Soil properties of the FE model.

Density (g/cm ³)	Elastic modulus (MPa)	Poisson's ratio	Friction angle (°)	Cohesion (kPa)
1983.00	68.000	0.290	21.49	0.842

Table 5
Parameters of modeling components involved in the study.

Component	Unit weight (kN/m ³)	Modulus (MPa)	Poisson's ratio	External diameter (m)	Lining thickness (m)
Segmental lining	25	36,500	0.38	12	0.45
Continuous lining	25	25,550	0.38	12	0.45

Step1: Train load generation using Simpack. A train speed of 80 km/h was applied. Each carriage has a base weight of 38 t, which increases by 12–24 t under maximum passenger loading. Within this range, 800 random carriage loads were sampled to represent different operational load conditions. For each case, the corresponding wheel-rail force time

history was generated in Simpack.

Step2: Damage modeling in Abaqus. Two types of lining damage were considered: voids behind segments and concrete spalling (See Fig. 12). Void damage was simulated by assigning zero mechanical properties (e.g., elastic modulus and Poisson's ratio) to a specific soil region behind the segment, representing loss of support. Concrete spalling was simulated geometrically through Boolean operations during model creation to remove the spalling part from the part geometry.

Step3: Simulation dataset generation in Abaqus. Structural dynamic responses were obtained by applying the wheel-rail forces to the tunnel-track FE model under different damage scenarios. The sample frequency is 200 HZ. Table 6 shows a summary of sample composition.

Step4: Output post-processing from Abaqus ODB files. Vertical acceleration responses were extracted from the ODB files at multiple monitoring points. To investigate the influence of sensor quantity, six monitoring layouts were considered, including 3, 6, 9, and 15 sensors along the tunnel lining. Fig. 13 shows the arrangements of monitoring points. Moreover, note that in the laboratory tests, a fixed vibration exciter was used to represent the train load, meaning that the “train” was assumed to remain within the monitored tunnel segment throughout the measurement period. To ensure consistency with this experimental setup, the simulated responses were trimmed accordingly: only the time period during which the entire train was fully inside the tunnel was retained for dataset generation.

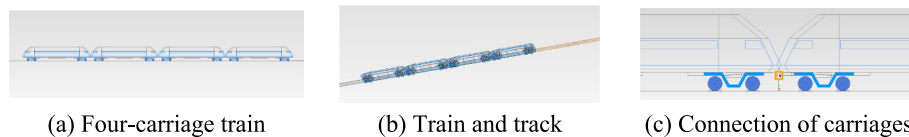


Fig. 10. Train-track subsystem.

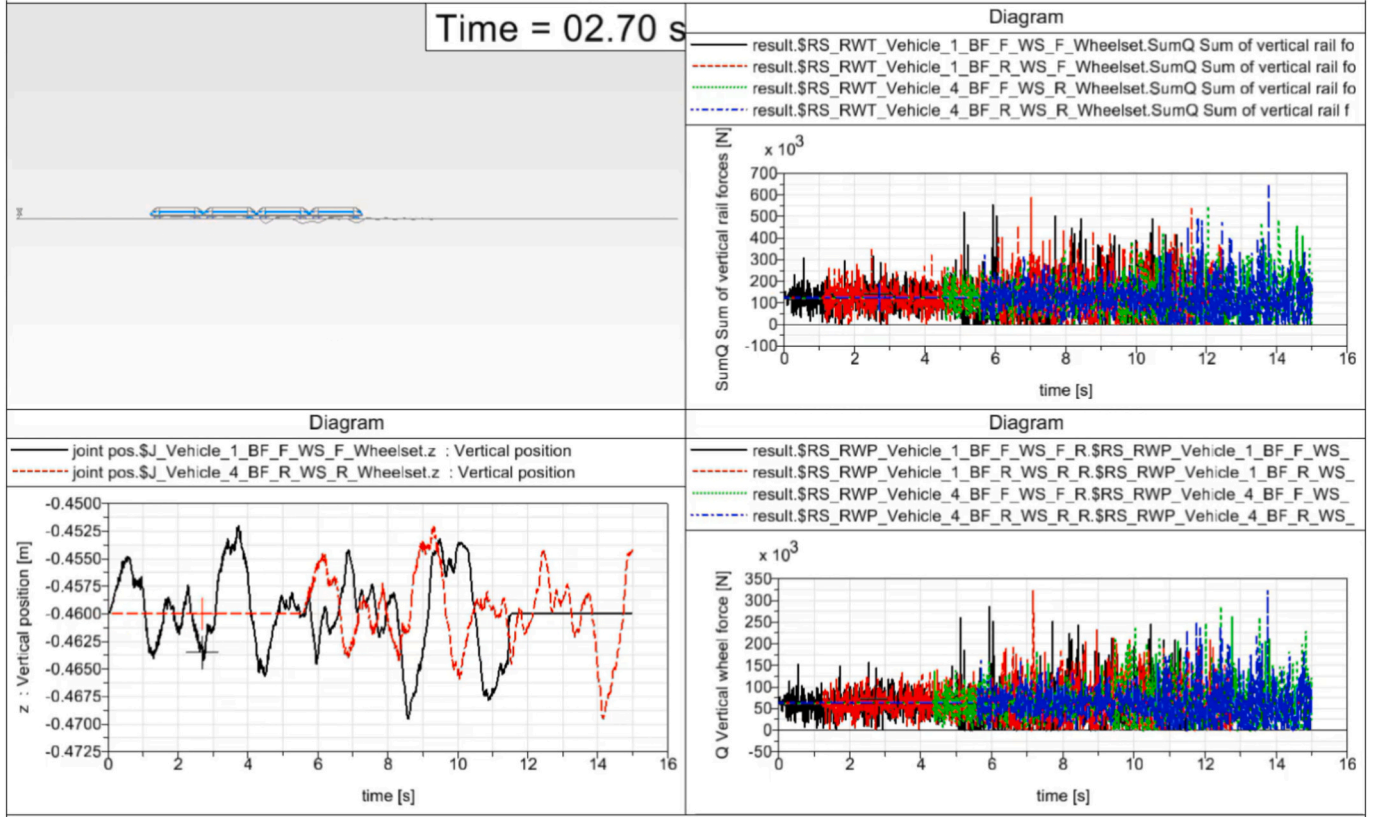


Fig. 11. Interaction forces between train and track.



Fig. 12. Damage geometries of void behind segment and concrete spalling.

Table 6

Simulation dataset (Damage detection zone = 10rings).

Simulation type	Segment ring	Location	level	Tran loads	Samples
Undamaged		–	–	800	1600 (Original + flip)
Void	10 rings	4	4	10	1600
Concrete spalling		3	1	40	1200

3.2. Case 1 void behind the segment

This subsection systematically evaluates the performance of the proposed framework in identifying and assessing voids behind the tunnel segment. The evaluation follows the two-stage workflow, beginning with model pretraining on numerical simulation data, and concluding with domain-adaptive model updating using experimental measurements.

3.2.1. Stage1: model pretraining

The pretraining stage aims to teach the network generalizable vibration-damage relationships using the simulation-based dataset for the void-behind-segment scenario. Each sample corresponds to one numerical simulation and contains a multi-sensor spectrogram tensor of shape $(S, 3, F, T)$, where each sensor provides magnitude, log-magnitude, and PSD channels computed from the STFT. The labels follow the exact hierarchical scheme defined in the dataset, including damage presence (0/1), void location (1 to 4), ring index mapped from physical rings R1 to R10 to network classes 1 to 10, and severity levels (1 to 4). Undamaged samples use class 0 for all three downstream tasks. In total, the void dataset consists of 3200 samples covering all void types and severities, along with undamaged cases for balanced training.

For time-frequency feature extraction, the raw acceleration signals were first cropped to retain the vibration response corresponding to the effective train-tunnel interaction period. Specifically, a relative time window of 0.31–2.01 s was selected with respect to the signal onset, ensuring consistency across samples. STFT was then applied using a Hann window with an FFT size of 64 and a hop length of 32, resulting in a 50% overlap between adjacent frames. To ensure uniform input

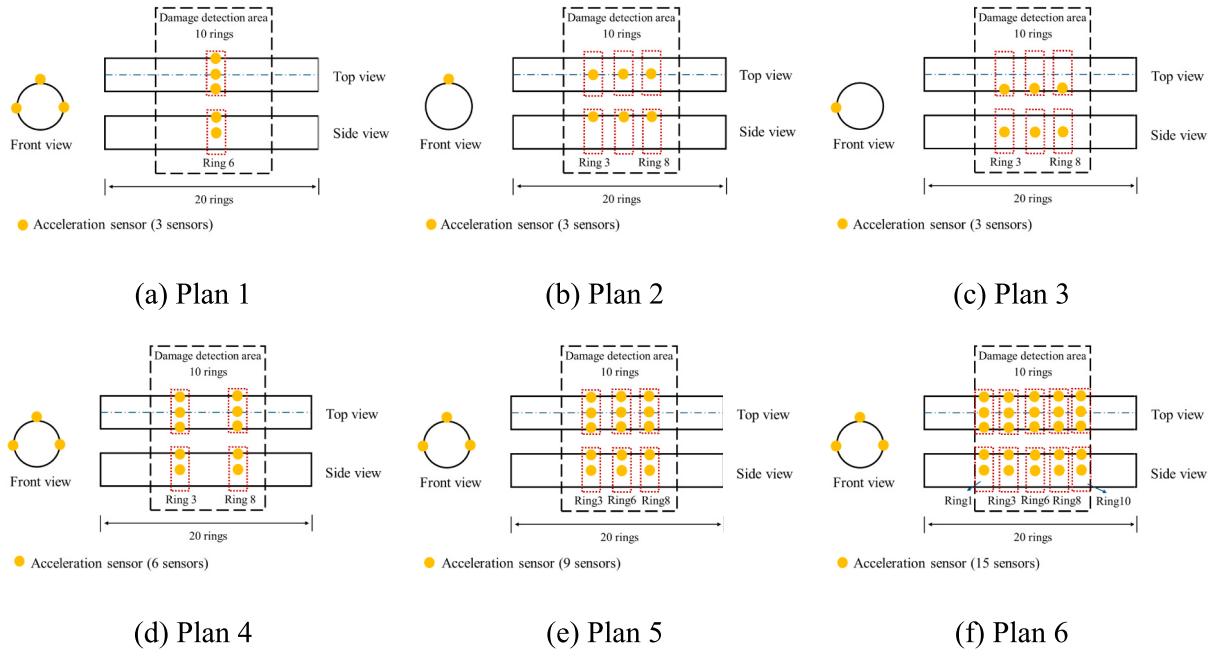


Fig. 13. Monitoring point layouts with 3, 6, 9, and 15 sensors.

dimensions for the neural network, the resulting spectrograms were center-cropped or zero-padded to a fixed length of 32 time frames. These settings provide a balanced trade-off between temporal resolution, frequency resolution, and computational efficiency, and were consistently used for both simulation and experimental datasets.

Before training, the full dataset is divided into training (60%), validation (20%), and testing (20%) subsets using a hierarchical stratification procedure. This ensures that every (ring, location, level) combination is proportionally represented across the splits. To explicitly prevent any risk of data leakage, this dataset split was executed at the independent simulation run (or experimental recording) level, prior to any sliding-window segmentation. This ensures that adjacent or overlapping time windows extracted from the same train passage do not inadvertently appear in both the training and testing sets. Furthermore, to address class imbalance without risking leakage, data augmentation techniques (such as signal flipping for the undamaged cases) were applied exclusively to the training subset after the data split was finalized. In this study, all networks were trained under identical conditions. These networks were optimized using Adam with a batch size of 8 and an initial learning rate of $0.5e-4$, scheduled by cosine annealing. Training was conducted for up to 100 epochs, with early stopping applied using a patience of 10 epochs based on validation performance. The model was trained on a computer with NVIDIA GTX 4060 (GPU) and Intel i9-13980HX (CPU).

To systematically justify the hyperparameter configurations of the proposed Trans-MTVFNet, a comprehensive sensitivity analysis was conducted on two critical groups of parameters: the multi-task loss weights and DA penalty weights. Table 7 presents the effect of varying multi-task weights on the overall testing accuracy. With the primary damage detection weight anchored at $w_d = 1$, the results exhibit a distinct inverted-U trend for each auxiliary task. The ring identification task requires the largest weight ($w_{ring} = 1.5$) to reach its optimal performance. Damage localization within a specific segment is a relatively straightforward spatial mapping task, effectively reaching its optimum at a much smaller weight ($w_{loc} = 1.1$). The severity assessment task, representing a moderate difficulty level, finds its optimal balance at $w_{level} = 1.2$. Further increasing any of these weights beyond their respective optimal points leads to a noticeable performance degradation due to gradient competition, where an overly dominant task suppresses

Table 7

Sensitivity analysis of multi-task loss weights.

$w_d = 1$					
$w_{ring} = 1, w_{level} = 1$		$w_{loc} = 1, w_{level} = 1$		$w_{loc} = 1, w_{ring} = 1$	
w_{loc}	Accuracy (%)	w_{ring}	Accuracy (%)	w_{level}	Accuracy (%)
0.1	97.33	0.1	92.67	0.1	96.24
0.3	97.7	0.3	93.88	0.3	96.37
0.5	97.92	0.5	95.68	0.5	96.9
0.7	98.06	0.7	96.81	0.7	97.27
0.8	98.12	0.8	97.57	0.8	97.4
0.9	98.16	0.9	98.08	0.9	97.91
1.0	98.29	1.0	98.29	1.0	98.29
1.1	98.61	1.1	98.6	1.1	98.45
1.2	98.37	1.2	98.87	1.2	99.07
1.3	98.28	1.3	99.12	1.3	98.58
1.5	98.23	1.5	99.43	1.5	98.35
1.7	98.2	1.7	98.91	1.7	97.57
1.9	98.11	1.9	98.18	1.9	97.25
Best weights		$w_{loc} = 1.1, w_{ring} = 1.5, w_{level} = 1.2$			

the feature representation space of others.

Table 8 demonstrates the sensitivity of the model to the DA penalty weights. The optimal cross-domain alignment is achieved at $\lambda_{global} = 0.1$ and $\lambda_{class} = 0.2$. If these weights are set too low (e.g., <

Table 8

Sensitivity analysis of DA penalty weights.

$\lambda_{global} = 0.1$		$\lambda_{class} = 0.2$	
λ_{class}	Accuracy (%)	λ_{global}	Accuracy (%)
0.05	96.04	0.05	97.33
0.1	98.74	0.1	98.67
0.2	98.94	0.2	97.93
0.3	98.44	0.3	94.51
0.5	97.51	0.5	94.41
Best weights		$\lambda_{global} = 0.1, \lambda_{class} = 0.2$	

0.05), the domain gap between the idealized simulation and the noisy experimental data remains insufficiently bridged. Conversely, if excessively high DA weights are applied, the network over-prioritizes domain confusion at the expense of task-specific classification. This causes the model to destroy the inherent discriminative damage features just to satisfy the domain discriminator, leading to severe negative transfer. The slightly higher requirement for λ_{class} (0.2) compared to λ_{global} (0.1) physically indicates that fine-grained, localized damage alignment necessitates slightly stronger regularization than macroscopic global alignment. Table 9 shows the proposed hyperparameters of Trans-MTVFNet. The selection of these hyperparameters was based on a combination of empirical tuning and reference to established practices in Transformer-based models.

To evaluate how the number of sensors affect the performance of the proposed multi-task vibration fusion network, we tested six monitoring point layouts (see Fig. 13). Table 10 shows performance summary results. Overall, a clear performance improvement can be observed as the number of sensing points increases, indicating that richer spatial vibration information significantly enhances model discrimination capability. For Plan 1–3 (3 sensors), the model achieves reasonable performance in damage presence detection, while the accuracy of ring number and damage localization remains limited. This suggests that sparse sensor layouts are sufficient for capturing global vibration anomalies but lack the spatial resolution required for fine-grained structural localization tasks. When the number of sensors increases to 6 and above (Plan 4–6), the model performance improves substantially across all tasks. In particular, Plan 4 (6 sensors) already achieves near-saturated performance, with precision values exceeding 99.70% for all four prediction tasks. Further increasing the number of sensors to 9 and 15 yields only marginal performance gains, while incurring a notable increase in training time, indicating diminishing returns beyond 6 sensors.

Among the four tasks, ring number classification consistently exhibits lower precision compared with damage presence and severity level, especially under sparse sensor layouts. From a structural dynamics perspective, detecting the overall presence of damage primarily relies on global structural responses (e.g., shifts in low-order modal frequencies). These global variations propagate extensively along the tunnel and can be easily captured by a few sensors. In contrast, pinpointing the exact damaged ring requires capturing localized, high-frequency vibration

Table 9
Hyperparameters of the proposed Trans-MTVFNet.

Model components	Name	Value
Shared ConvNet Encoder	Input channels	3
	Input size	224×224
	Number of layers	6
Sensor Embedding	Output embedding dimension d	256
	Number of sensors S	3 / 6 / 9 / 12 / 15
	Positional embedding dimension	256
Transformer Encoder	Embedding dimension	256
	Number of attention heads	3 / 4
	Feedforward dimension	256
Training setup	Damage loss	Focal loss ($\alpha = 0.5$, $\gamma = 2$)
	No-damage weak supervision weight	$w_{\text{weight}} = 0.2$
	Penalty for non-zero class in damaged samples	$\lambda_{\text{forbid0}} = 0.1$
	Task weights	$w_d = 1$;
		$w_{\text{loc}} = 1.1$;
		$w_{\text{ring}} = 1.5$;
		$w_{\text{level}} = 1.2$
(Only for stage 2)	Global DA	$\lambda_{\text{global}} = 0.1$
	Class-conditional DA	$d_{\text{hidden}} = 128$ $\lambda_{\text{class}} = 0.2$ $m_{\text{dgamma}} = 1$

Table 10

Model performance results on the training & testing dataset.

Layout	Training & validation stage		Weighted average F1-score of Testing stage (Class number)			
	Best train loss	Training time	Task 1 Damage presence (2)	Task 2 Ring number (10)	Task 3 Damage location (4)	Task 4 Severity level (4)
Plan 1	0.0170	1.5 h	81.24%	72.06%	79.71%	82.65%
Plan 2	0.0145	2.1 h	86.94%	83.27%	86.32%	86.03%
Plan 3	0.0286	1.8 h	77.04%	73.78%	76.75%	78.33%
Plan 4	0.0052	4.4 h	99.84%	99.70%	99.80%	99.89%
Plan 5	0.0064	9.6 h	99.79%	99.68%	99.86%	99.85%
Plan 6	0.0053	18.0 h	99.87%	99.69%	99.85%	99.88%

features induced by the defect. Because high-frequency stress waves attenuate rapidly within the concrete lining and dissipate into the surrounding soil, sparse sensor arrays lack the necessary spatial resolution to capture these local features, inevitably leading to lower localization accuracy. To improve the accuracy of ring number classification, Stage 2 introduces a DA strategy via transfer learning. This process preserves the global vibration knowledge learned from simulations while adapting the network to realistic wave propagation characteristics and sensor noise patterns, thereby specifically enhancing the robustness of ring number localization under sparse sensing conditions.

3.2.2. Test data processing

To prepare the physical-twin measurements for model updating and evaluation, a consistent data-processing pipeline was applied to all experimental vibration data. Fig. 14 illustrates a representative raw acceleration signal collected from one case. The processing pipeline for experimental test data is kept consistent with that used for simulation data to ensure compatibility during transfer learning. The main differences are related to data acquisition conditions. Specifically, the raw experimental acceleration signals are originally sampled at 2000 Hz and are first downsampled to 200 Hz to match the simulation setting and reduce redundancy. To ensure data quality and eliminate transient disturbances, the first 5 s and last 5 s of each recording are discarded. Following the same splitting protocol to prevent data leakage, the remaining signals are then segmented using the same strategy as in the simulation data, where a window length of 1.7 s is treated as one sample for subsequent time-frequency transformation and model input generation.

To further clarify the inherent domain discrepancy between the numerical simulations and laboratory tests, a comparative analysis of the signal characteristics was conducted across six distinct measurement points (Fig. 15). The acceleration waveforms from both domains were first normalized to eliminate absolute scale disparities, followed by the computation of their PSD. As illustrated in the overlaid PSD comparison, the semi-transparent bounds of individual measurement points and their corresponding mean curves reveal a consistent trend: while the numerical simulation reliably captures the fundamental low-frequency dynamics of the structure, a conspicuous mismatch exists across the broader frequency spectrum. Specifically, the experimental data systematically exhibits richer, localized high-frequency energy components and environmental complexities that the idealized numerical model fundamentally fails to replicate. This universally observed spectral gap visually and quantitatively highlights the significant feature distribution mismatch between the source domain (simulation) and target domain (experiment), thereby firmly justifying the necessity of introducing the proposed DA strategy to effectively bridge this gap.

Each sample converted into a time-frequency representation using STFT. From each complex spectrum $A(f, t)$, three complementary channels, magnitude $|A|$, log-magnitude $\log(|A| + \epsilon)$, and power spectral density $|A|^2$, were extracted and stacked into a 3-channel spectrogram of

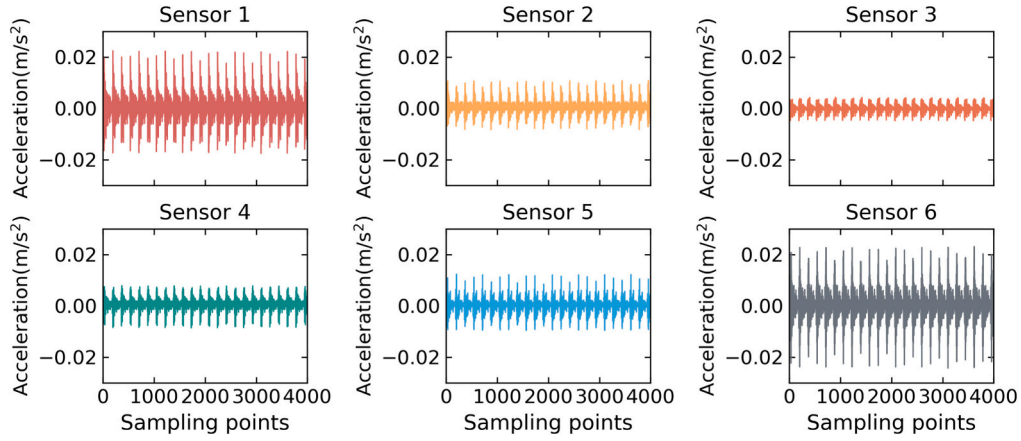


Fig. 14. Raw acceleration signal in the void test.

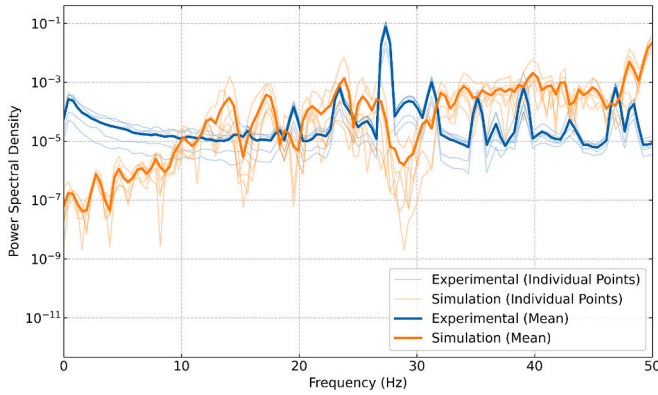


Fig. 15. Comparison of PSD between numerical simulations and experimental measurements.

shape $S \times 3 \times F \times T$, matching the simulation data format. Due to experimental duration and sensor relocation constraints, only five ring positions (R4 to R8) were tested in the physical setup. Accordingly, the ring-level label space was reduced to six classes (0 for no-damage, 1–5 for R4–R8), and the ring-prediction head of the model was adjusted to output:

Linear($d \rightarrow 6$).

After label compression and spectrogram generation, all experimental samples were normalized. These processed samples formed the experimental dataset, which is used in the following section to update the model toward the real tunnel response.

3.2.3. Stage2: Model updating

After pretraining the multi-task vibration fusion network on numerical simulation data, the model is further adapted to real experimental measurements through a DT-driven DA strategy. This section systematically evaluates the model updating behavior from three complementary perspectives: (1) the influence of experimental data quantity on adaptation performance, (2) an ablation study of different DA strategies, and (3) performance comparisons with alternative benchmark models.

Similar to the simulation dataset, the experimental dataset contains a total of 3200 samples, including 1600 undamaged samples and 1600 damaged samples. In practice, damaged data are much more difficult to obtain in real tunnel environments. Therefore, all undamaged samples from the experiments are retained, and the impact of using different proportions of damaged samples on the model performance is systematically investigated in this study. To quantify how the number of real

measurements affect adaptation performance, five training subsets were created using 0%, 20%, 40%, 60%, 80%, and 100% of the damaged experimental samples, while the test set remained fixed.

Fig. 16 shows the influence of the amount of damaged experimental samples on model performance, with and without simulation-based pretraining. Model performance consistently improves as the proportion of damaged experimental samples increases, indicating that real measurements play a critical role in effective damage recognition and evaluation. When sufficient experimental data are available, both models achieve comparable and stable performance across all evaluation metrics. This suggests that, given enough real data, the model can learn task-relevant representations directly from the target domain, reducing its dependence on pretrained initialization. In contrast, under limited experimental data conditions, the model with simulation-based pretraining demonstrates consistently better performance than the model trained from scratch. However, the performance gain remains moderate rather than substantial. This observation implies that while pretraining on simulated data provides a useful initialization and facilitates early-stage learning, notable feature discrepancies exist between simulated and experimental data. As a result, the transferable knowledge obtained from simulation contributes only partially to the final performance, whereas experimental data provide more direct and domain-relevant information for damage characterization. Overall, these results indicate that simulation data can support feature learning when real data are scarce, but experimental measurements remain essential for achieving robust and accurate DT adaptation.

To evaluate the contribution of each algorithmic contribution to the

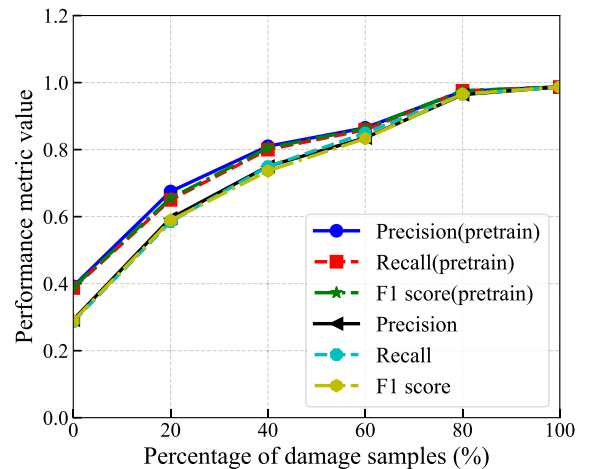


Fig. 16. Effect of damaged sample proportion on adaptation performance.

overall performance, a comprehensive ablation study was conducted. Five core architectural components and techniques were defined and compared: (1) Pretraining on an external dataset; (2) Global DA for global feature alignment; (3) Class-wise DA for category-specific feature alignment; (4) Multi-task learning (MTL) design (as opposed to single-task learning, STL); and (5) Gated inference mechanism. Table 11 presents the weighted average F1-score results of testing stage evaluating the contributions of different strategies. Without pretraining or DA, the model achieves the lowest performance, indicating limited generalization when trained directly on experimental data. Introducing simulation-based pretraining leads to consistent but moderate performance improvements across all tasks. This suggests that pretrained representations from simulation provide a useful initialization; however, the gains remain limited due to inherent feature discrepancies between simulated and experimental data. Further incorporating DA results in more substantial performance improvements. Global DA improves overall accuracy across tasks by reducing coarse domain shifts, while class-wise DA yields more pronounced gains, particularly for tasks requiring fine-grained structural discrimination. Notably, the ring number classification task (Task 2) benefits the most from DA. Compared to the pretraining-only model, class-wise DA improves Task 2 accuracy from 93.88% to 95.89% (+2.01%), representing the largest relative gain among all tasks. This result directly supports the motivation for introducing DA to enhance the reliability of ring-level damage identification under cross-domain conditions. The best overall performance is achieved when both global and class-wise DA strategies are jointly applied. Under this configuration, all tasks reach their highest accuracy, with consistent improvements of approximately 2–3% compared to models without DA. These results demonstrate that simulation-based pretraining provides a beneficial foundation, while effective DA, particularly class-wise alignment, is essential for robust multi-task damage evaluation and accurate ring-level damage assessment.

As shown in Table 11, the results indicate that STL achieves sub-optimal performance, particularly suffering a significant drop in the challenging ring number identification task (e.g., dropping to 94.15%). From a structural dynamics perspective, these damage-related tasks are intrinsically coupled, as they all depend on the underlying physics of wave scattering and energy dissipation. The proposed multi-task design capitalizes on this physical correlation by allowing the auxiliary tasks to act as implicit regularizers. Moreover, the model without the gated mechanism suffers severe degradation, performing even worse than STL. This degradation is primarily caused by gradient contamination: forcing the network to “locate” damage in a healthy signal generates erroneous, conflicting loss gradients. These meaningless gradients backward-propagate and severely pollute the shared feature space, dragging down the accuracy of all tasks. By integrating both the multi-task architecture and the physical logic gate, the proposed Trans-MTVFNet prevents gradient contamination while fully exploiting task synergy, ultimately yielding the highest evaluation metrics.

To evaluate the effectiveness of the proposed Trans-MTVFNet, a comparative study is conducted against several representative network

architectures, including CNN-based models (EfficientNetV2, ResNet34, RegNetY-8) [26–28] and transformer-based models (ViT and Swin Transformer) [29,30]. All models are trained and evaluated under the same experimental settings, and their performance is comprehensively evaluated using weighted average Precision, Recall, and F1-score across the four damage-related tasks, as reported in Table 12. Figs. 17, 18, and 19 show confusion matrices on the testing dataset for Resnet34, Swin Transformer, and Trans-MTVFNet. Overall, the proposed Trans-MTVFNet consistently achieves the best or highly comparable performance across all tasks. In particular, Trans-MTVFNet attains the highest Precision, Recall, and F1-scores on Task 1 (damage presence) and Task 2 (ring number), while delivering highly competitive results on Task 3 (damage location) and Task 4 (severity level) that are nearly identical to the Swin Transformer. The architectural advantage of Trans-MTVFNet is especially evident in the challenging ring number classification task (Task 2). Conventional CNN-based models exhibit a significant performance degradation, particularly characterized by a sharp drop in Recall (e.g., RegNetY_008 drops to 79.36%). This indicates their vulnerability to missing subtle, high-frequency damage signatures. In contrast, Trans-MTVFNet outperforms conventional CNN-based models by a large margin and achieves performance comparable to Swin Transformer. This demonstrates its strong capability in capturing fine-grained structural dependencies and localized wave scattering features.

From a computational perspective, Trans-MTVFNet demonstrates a favorable balance between accuracy and efficiency. While transformer-based models such as ViT and Swin Transformer achieve strong performance, they require significantly longer training times. In contrast, Trans-MTVFNet reduces training time by more than 50% compared to Swin Transformer, while maintaining nearly identical performance across all tasks. Compared with CNN-based models, Trans-MTVFNet achieves substantially higher accuracy at a moderate computational cost. These results indicate that the proposed Trans-MTVFNet effectively combines the strengths of convolutional and transformer-based architectures, achieving superior accuracy with improved computational efficiency. This balanced performance makes it particularly suitable for practical DT applications, where both accuracy and training efficiency are critical considerations.

3.3. Case 2 concrete spalling

This subsection investigates the framework's capability to detect and evaluate concrete spalling defects. Consistent with the first case, the assessment procedure is divided into simulation-based model pretraining, experimental data collection, and subsequent domain-adaptive model updating to bridge the simulation-to-reality gap.

3.3.1. Stage1: model pretraining

In Case 2, simulation-based pretraining is conducted for the concrete spalling scenario following the same general pipeline as described in Case 1. To reduce the computational cost associated with numerical model-based data generation, the simulation setup is simplified by considering only a single depth of concrete spalling damage. Therefore,

Table 11
Ablation study of different strategies.*

Ablation setting					Task 1 Damage Presence	Task 2 Ring Number	Task 3 Damage Location	Task 4 Severity Level
Pretraining	Global DA	Class-wise DA	MTL	Gated				
×	×	×	Δ	Δ	96.51%	93.18%	96.46%	96.55%
Δ	×	×	Δ	Δ	97.24%	93.88%	97.20%	97.11%
Δ	Δ	×	Δ	Δ	97.48%	94.32%	97.32%	97.15%
Δ	×	Δ	Δ	Δ	98.85%	95.89%	98.92%	98.83%
Δ	Δ	Δ	×	–	98.52%	94.15%	96.84%	96.25%
Δ	Δ	Δ	Δ	×	97.14%	91.68%	93.42%	94.75%
Δ	Δ	Δ	Δ	Δ	99.95%	98.98%	99.21%	98.92%

* Δ: means using corresponding module; ×: means without the corresponding module; –: means not applicable (STL does not require gating).

Table 12
Comparison of testing accuracy for different networks.

Networks	EfficientNetV2	Resnet34	RegNetY_008	ViT	Swin Transformer	Trans-MTVFNet
Task 1 Damage Presence						
Precision	97.50%	98.91%	98.28%	99.38%	99.69%	99.84%
Recall	97.50%	98.91%	98.28%	99.38%	99.69%	99.84%
F1-score	97.50%	98.91%	98.28%	99.38%	99.69%	99.84%
Task 2 Ring Number						
Precision	89.15%	86.10%	81.12%	95.15%	98.44%	99.07%
Recall	88.78%	86.03%	79.36%	94.97%	98.43%	99.06%
F1-score	88.96%	86.05%	80.23%	95.06%	98.43%	99.06%
Task 3 Damage Location						
Precision	95.62%	92.86%	92.35%	96.95%	99.07%	99.06%
Recall	95.51%	92.73%	92.06%	96.85%	99.06%	99.38%
F1-score	95.56%	92.72%	92.20%	96.90%	99.06%	99.22%
Task 4 Severity Level						
Precision	96.55%	93.53%	91.20%	97.85%	98.76%	98.45%
Recall	96.47%	93.34%	90.79%	97.80%	98.74%	98.43%
F1-score	96.51%	93.37%	90.99%	97.82%	98.75%	98.44%
Training time	6.5 h	1.8 h	1.1 h	7.2 h	9.4 h	3.9 h

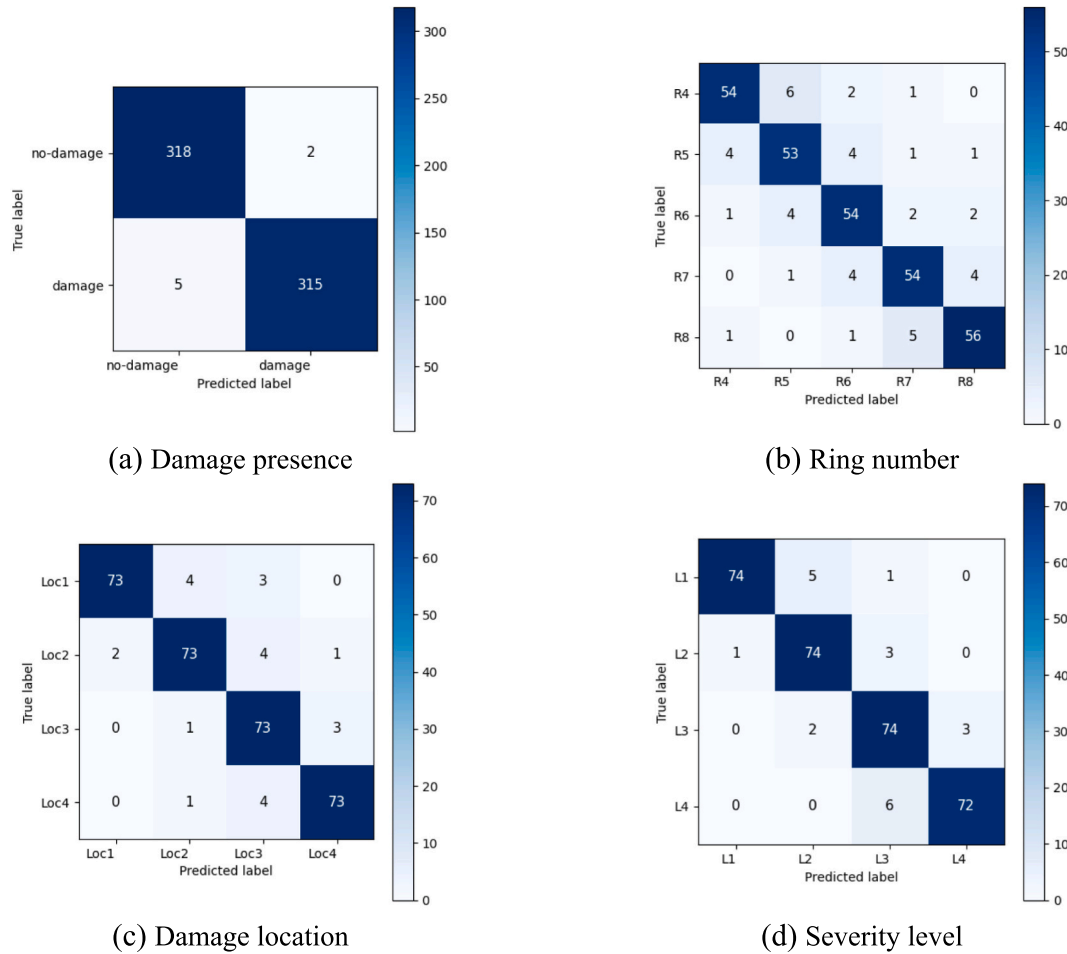


Fig. 17. Confusion matrices on the testing dataset for Resnet34.

three damage-related tasks are included in this case study: damage presence, ring number identification, and damage localization. The performance of different pretraining layouts is summarized in Table 13. Overall, increasing the training complexity leads to clear performance improvements across all tasks, particularly for ring number

identification and damage localization. While simpler training plans show limited accuracy, more comprehensive pretraining configurations achieve F1-scores exceeding 98% for damage presence and localization, and above 96% for the ring number task. Further increasing training time yields only marginal performance gains, indicating diminishing

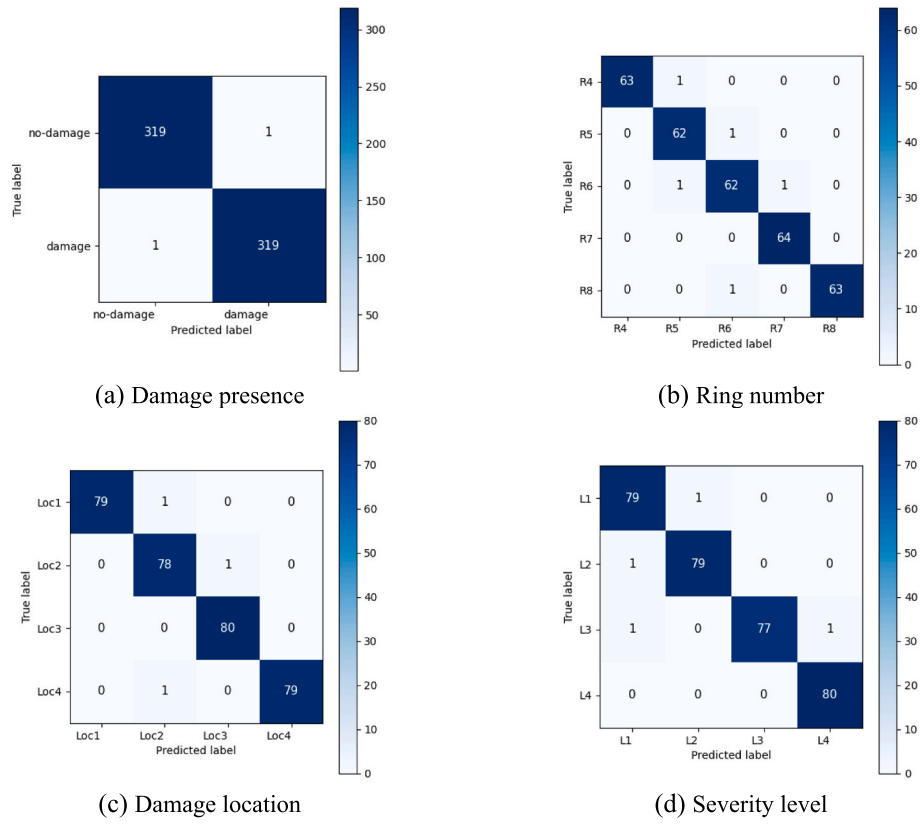


Fig. 18. Confusion matrices on the testing dataset for Swin Transformer.

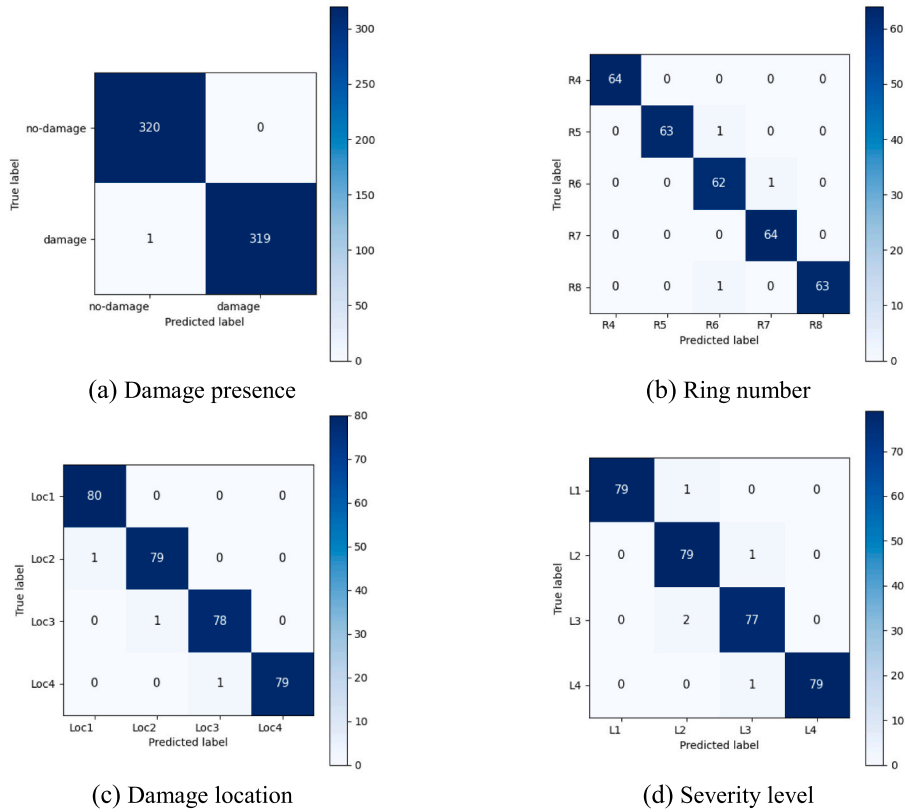


Fig. 19. Confusion matrices on the testing dataset for Trans-MTVFNet.

returns. Therefore, Plan 4 is selected to provide a robust initialization for subsequent model updating in Stage 2, achieving high accuracy with

Table 13

Model performance results on the training & testing dataset.

Layout	Training & validation stage		Weighted average F1-score of Testing stage (Class number)		
	Best train loss	Training time	Task 1	Task 2	Task 3
			Damage presence (2)	Ring number (10)	Damage location (4)
Plan 1	0.0412	0.8 h	73.24%	69.87%	77.09%
Plan 2	0.0378	1.1 h	83.34%	79.14%	85.67%
Plan 3	0.0486	0.9 h	80.61%	72.07%	79.38%
Plan 4	0.0083	3.2 h	98.54%	96.32%	98.54%
Plan 5	0.0078	7.1 h	98.39%	97.03%	98.36%
Plan 6	0.0081	12.3 h	98.47%	97.11%	98.25%

reasonable computational cost.

3.3.2. Test data collection

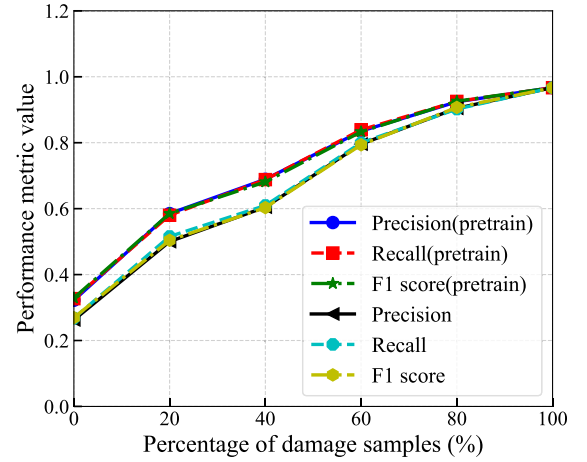
Fig. 20 shows representative raw acceleration signals recorded by multiple sensors during the concrete spalling test. Similar to the void dataset in Section 3.2.2, the same data-processing pipeline was applied to vibration measurements in the concrete spalling test. After data processing, label adjustment and spectrogram generation, all experimental samples are normalized to form the experimental dataset, which is used in the following section for model updating toward real tunnel responses.

3.3.3. Stage2: Model updating

This section also investigates the model updating behavior from three complementary perspectives: (1) the influence of experimental data availability on adaptation performance, (2) the effectiveness of different DA strategies through ablation analysis, and (3) performance comparisons with representative benchmark models.

Fig. 21 shows effect of damaged sample proportion on adaptation performance. As shown in Fig. 21, model performance consistently improves as the proportion of damaged experimental samples increases. When a sufficient proportion of damaged experimental data is available, the performance gap between models with and without simulation-based pretraining becomes marginal. In contrast, when damaged experimental samples are limited, simulation-based pretraining provides a more stable starting point for model updating. The pretrained model exhibits faster performance improvement and more reliable predictions in low-data regimes. This behavior reflects the partial alignment between simulated and experimental features: pretraining facilitates adaptation but cannot fully substitute for real measurements.

Table 14 presents the ablation study results for Case 2. Compared with the void case, the overall performance gains obtained in the concrete spalling scenario are relatively more modest. This difference can be

**Fig. 21.** Effect of damaged sample proportion on adaptation performance.

attributed to the physical characteristics of concrete spalling damage, which typically involves smaller affected areas and more localized deterioration. As a result, the induced variations in acceleration signals are less distinctive and more difficult to discriminate, particularly under experimental noise and environmental interference. Nevertheless, the performance trends across all model variations align perfectly with the theoretical analysis established in Case 1. Specifically, the progressive addition of DA (especially Class-wise DA) substantially bridges the complex domain shifts. More importantly, the critical roles of the multi-task and gated designs are highly reaffirmed in this challenging scenario. Reverting to STL causes a noticeable drop in ring identification accuracy (down to 93.15%), while removing the physical logic gate leads to an even more severe performance collapse across all metrics due to the aforementioned gradient contamination. Therefore, by integrating all structural innovations, the full Trans-MTVFNet consistently attains the highest accuracy, demonstrating its robustness and transferability in identifying subtle structural defects.

Table 15 compares the proposed Trans-MTVFNet with several representative CNN-based and transformer-based architectures in terms of comprehensive testing metrics (Precision, Recall, and F1-score) and training efficiency. Figs. 22, 23, and 24 show confusion matrices on the testing dataset for Resnet34, Swin Transformer, and Trans-MTVFNet. Overall, Trans-MTVFNet achieves the highest or highly competitive performance across all three tasks, demonstrating robust generalization capability for spalling damage identification. Compared with conventional CNN models, the proposed network provides clear performance improvements, especially for the challenging ring number classification (Task 2). Because concrete spalling induces highly subtle and localized vibration changes, CNNs struggle to capture these weak signatures. In

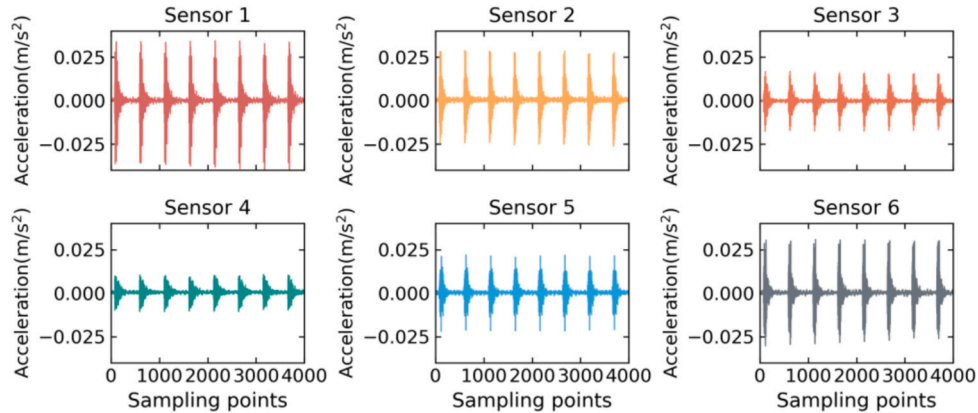
**Fig. 20.** Raw acceleration signal in the concrete spalling test.

Table 14
Ablation study of different strategies.*

Ablation setting					Task 1 Damage Presence	Task 2 Ring Number	Task 3 Damage Location
Pretraining	Global DA	Class-wise DA	MTL	Gated			
×	×	×	Δ	Δ	94.23%	89.69%	91.65%
Δ	×	×	Δ	Δ	95.41%	89.98%	93.04%
Δ	Δ	×	Δ	Δ	95.69%	92.27%	94.72%
Δ	×	Δ	Δ	Δ	97.37%	94.85%	95.29%
Δ	Δ	Δ	×	–	97.85%	93.15%	95.50%
Δ	Δ	Δ	Δ	×	96.55%	90.45%	92.80%
Δ	Δ	Δ	Δ	Δ	99.13%	96.91%	97.13%

* Δ: means using corresponding module; ×: means without the corresponding module; –: means not applicable (STL does not require gating).

Table 15
Comparison of testing precision for different networks.

Networks	EfficientNetV2	Resnet34	RegNetY_008	ViT	Swin Transformer	Trans-MTVFNet
Task 1 Damage Presence						
Precision	96.25%	97.71%	97.50%	98.75%	98.96%	99.38%
Recall	96.25%	97.71%	97.50%	98.75%	98.96%	99.38%
F1-score	96.25%	97.71%	97.50%	98.75%	98.96%	99.38%
Task 2 Ring Number						
Precision	92.19%	91.17%	86.68%	94.49%	96.78%	96.75%
Recall	92.19%	91.06%	86.73%	94.51%	96.64%	96.66%
F1-score	92.19%	91.08%	86.70%	94.50%	96.68%	96.67%
Task 3 Damage Location						
Precision	94.75%	94.90%	93.18%	96.05%	97.06%	97.91%
Recall	94.78%	94.89%	93.16%	96.03%	97.06%	97.91%
F1-score	94.76%	94.89%	93.17%	96.04%	97.06%	97.90%
Training time	5.9 h	1.3 h	0.9 h	6.5 h	7.3 h	3.1 h

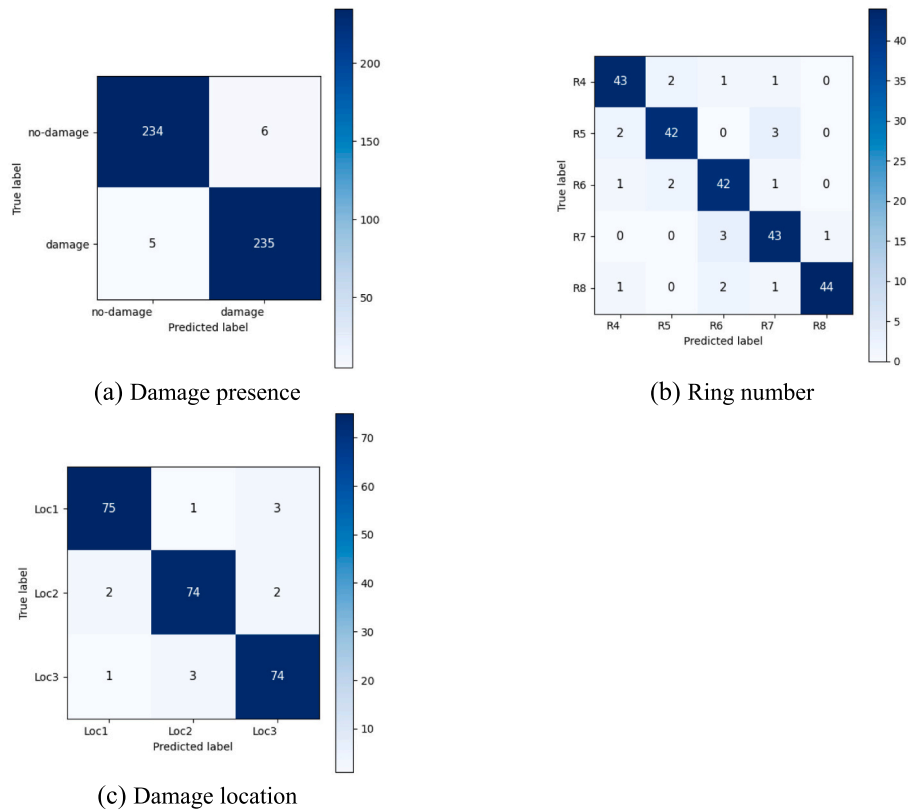


Fig. 22. Confusion matrices on the testing dataset for Resnet34.

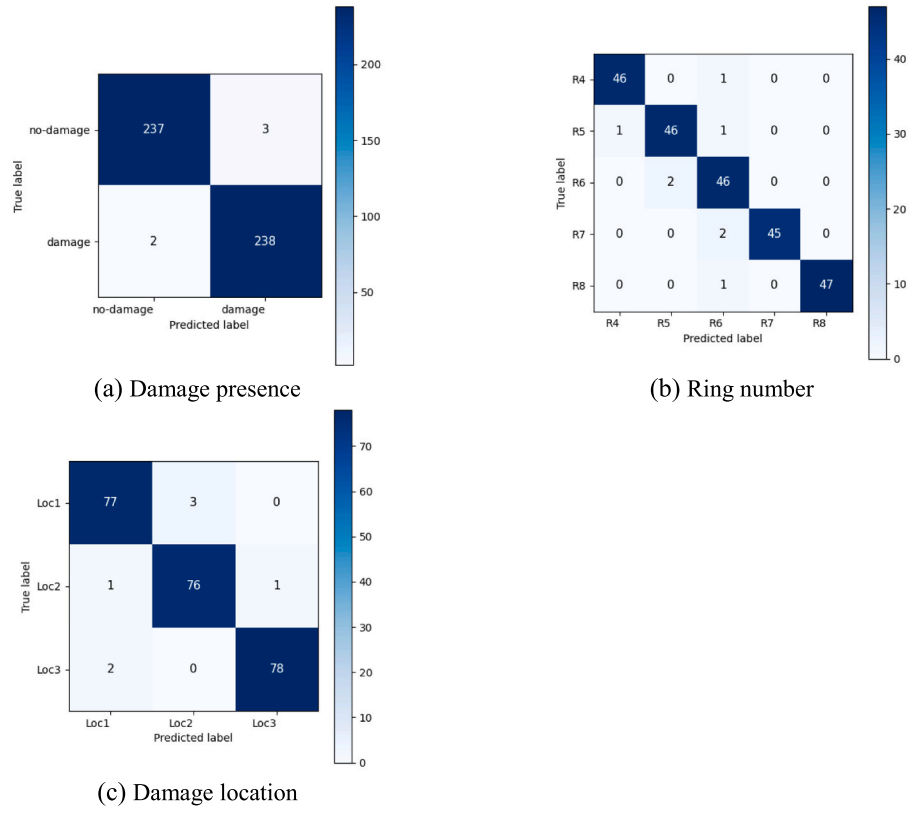


Fig. 23. Confusion matrices on the testing dataset for Swin Transformer.

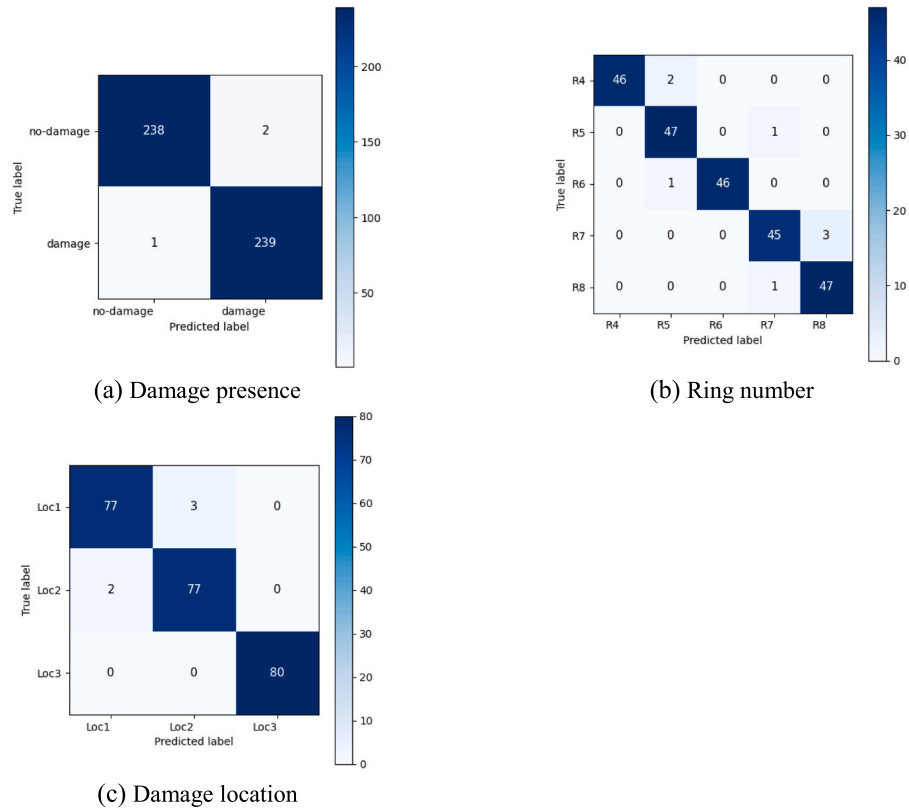


Fig. 24. Confusion matrices on the testing dataset for Trans-MTVFNet.

contrast, Trans-MTVFNet effectively overcomes this limitation, maintaining a robust and balanced precision and recall through its enhanced feature fusion and long-range dependency modeling. Furthermore, while transformer-based models such as ViT and Swin Transformer deliver competitive accuracy, they incur substantially higher computational costs. Trans-MTVFNet requires only 3.1 h of training time, less than half the time of the Swin Transformer (7.3 h), while delivering comparable or superior multi-task accuracy. These results indicate that Trans-MTVFNet achieves a highly favorable balance between multi-task detection accuracy and training efficiency, making it practically well-suited for continuous model updating in digital twin-based tunnel damage assessment.

In summary, as demonstrated in the comprehensive case studies involving both void and concrete spalling defects, the proposed Trans-MTVFNet successfully achieved high-precision, multi-task damage assessment on the laboratory-scale physical testbed. While the experimental results validate the good performance of Trans-MTVFNet, its practical applicability within a DT framework is fundamentally underpinned by its robustness, generalization capability, and computational efficiency. In terms of robustness, the model inherently prevents logical error propagation through its task-coupled gating strategy, which conditionally activates fine-grained assessment heads only when damage is explicitly detected, thereby naturally filtering out false positives induced by unmodeled environmental noise. Furthermore, the framework demonstrates superior generalization across diverse structural conditions through a dual-strategy DA module. By aligning both global response distributions and class-conditional features, the model can transfer generic damage representations pre-trained on numerical simulations to physical tunnel domains with differing boundary conditions and material properties. Finally, a highly favorable computational efficiency is ensured in this architecture by relying on an optimized ConvNeXt backbone for the extraction of high-dimensional time-frequency features. This effectively compresses the spatial-spectral features, allowing the subsequent Transformer encoder to focus purely on cross-sensor fusion without the quadratic computational bottleneck of standard Vision Transformers. As a result, Trans-MTVFNet reduces training time by over 50% compared to equivalent models like the Swin Transformer, while consistently delivering superior multi-task accuracy.

4. Conclusions

This paper presented a domain-adaptive digital twin (DT) framework for vibration-based damage detection and assessment of shield tunnels. By integrating simulation-based pretraining, an efficient multi-task transferable network (Trans-MTVFNet), and domain adaptation-driven model updating, the proposed framework provides a promising framework for vibration-based monitoring of shield tunnels. The main conclusions of this paper can be summarized as follows:

- (1) Simulation-based pretraining provides a useful foundation. Pretraining on numerical simulation data enables the model to learn general damage-related representations and offers a stable initialization, particularly when experimental damaged data are scarce. However, the direct contribution of simulated features is constrained by domain discrepancies.
- (2) Incorporating domain adaptation strategies improves model performance when transferring from simulation to experimental data. In particular, class-wise domain adaptation plays a critical role in enhancing tasks that require fine-grained structural discrimination, such as ring-level damage identification.
- (3) The proposed convolutional framework achieves a favorable accuracy-efficiency balance. Compared with transformer-based and conventional CNN-based benchmarks, the proposed model delivers competitive or superior accuracy while substantially reducing training time, making it suitable for DT applications.

- (4) Compared with void damage, concrete spalling presents more subtle vibration signatures due to its smaller affected area, leading to relatively lower performance gains. This highlights the importance of task-aware adaptation strategies for different damage mechanisms.

Despite encouraging results, several limitations and deployment challenges remain before transitioning this framework to real-world tunnel environments. First, the numerical models use simplified damage configurations that may not fully capture the complexity of actual deterioration. Second, the current framework operates strictly under a supervised DA setting, requiring labeled experimental data for model updating, and does not yet consider unsupervised or semi-supervised scenarios for unlabeled target data. Third, current validations rely on controlled laboratory data. Real-world deployment faces severe data scarcity in the target domain. Acquiring comprehensive data on critical structural damage in operational tunnels is inherently difficult, leading to imbalanced datasets. Operational tunnels also introduce complex data anomalies and environmental noise (e.g., traffic-induced vibrations and electromagnetic interference) that differ significantly from controlled laboratory conditions, potentially degrading feature extraction. For continuous, real-time deployment on edge-computing devices, the proposed algorithm will require further optimization and pruning.

Future research will focus on extending the proposed framework in several directions. More realistic and diverse simulation scenarios will be incorporated to better reflect complex damage evolution. Then, future studies will explore a phased domain adaptation strategy. This approach will utilize expansive numerical models to generate baseline training data, leverage the laboratory experimental data as a transitional domain-adaptation layer, and ultimately fine-tune the algorithm using limited operational data. This deployment roadmap will bridge the gap between theoretical algorithm design and practical structural health monitoring in complex engineering environments. In addition, semi-supervised or self-supervised learning strategies will be investigated to reduce dependence on labeled damaged data. Finally, real-time model updating and long-term DT deployment will be explored to support continuous tunnel condition monitoring and maintenance decision-making.

CRedit authorship contribution statement

Xiangyu Chang: Conceptualization, Methodology, Software, Validation, Writing - original draft. **Hongyun Fan:** Conceptualization, Methodology, Software, Validation, Writing - review & editing. **Bozheng Yang:** Conceptualization, Methodology, Software, Writing - review & editing. **Yuguang Fu:** Supervision, Funding acquisition, Methodology, Investigation, Writing - review & editing. **Hao Wang:** Supervision, Investigation, Writing - review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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